Radiation-hydrodynamic phenomena: Self-shadowing and Vertical Shear Instability

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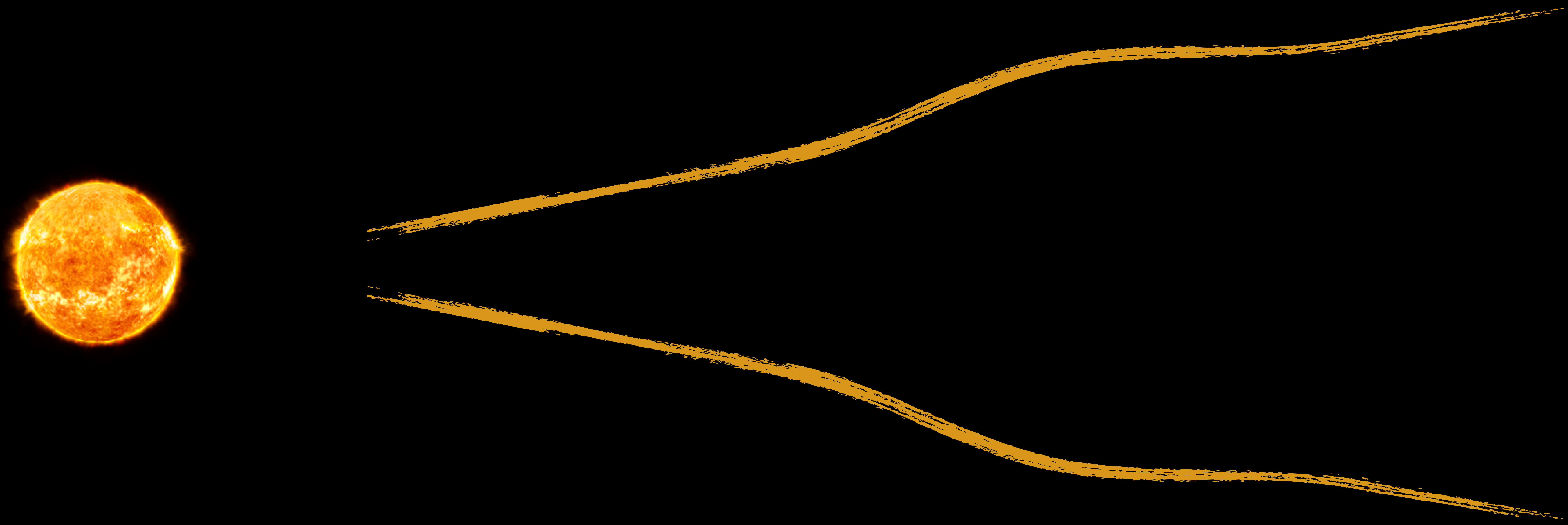
In collaboration with **Dhruv Muley, Prakruti Sudarshan, Mario Flock,** and **Hubert Klahr** (MPIA)

PFITS+ meeting 2026

Image: NASA/JPL-Caltech

SELF-SHADOWING INSTABILITY?



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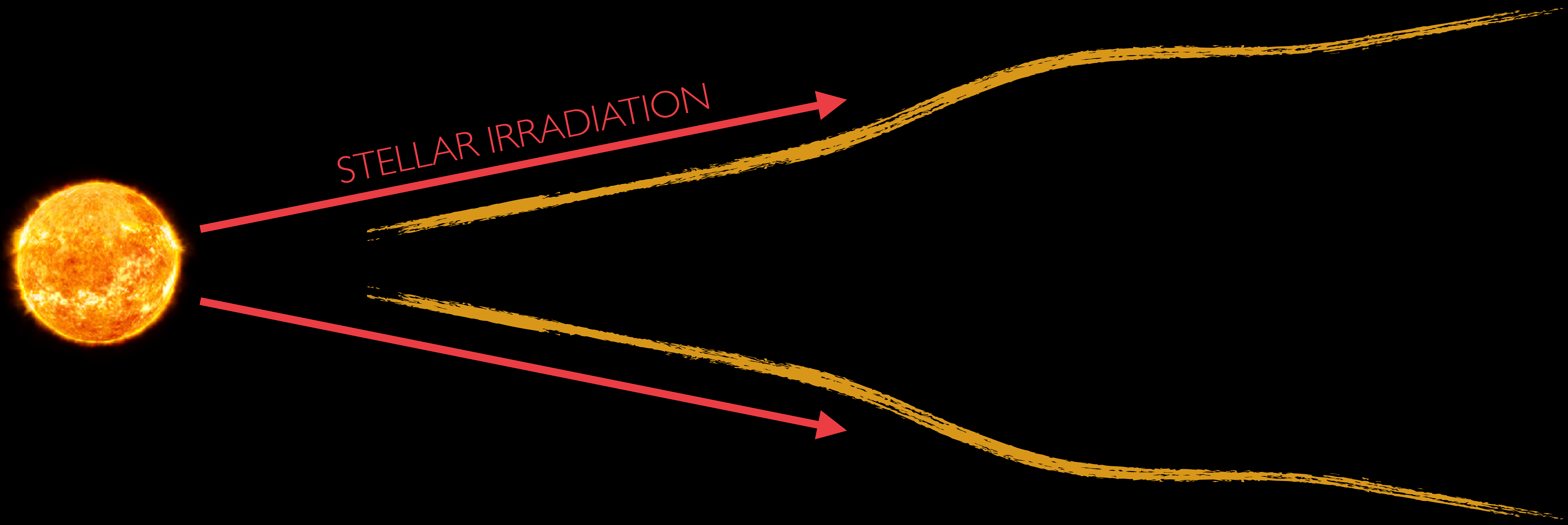

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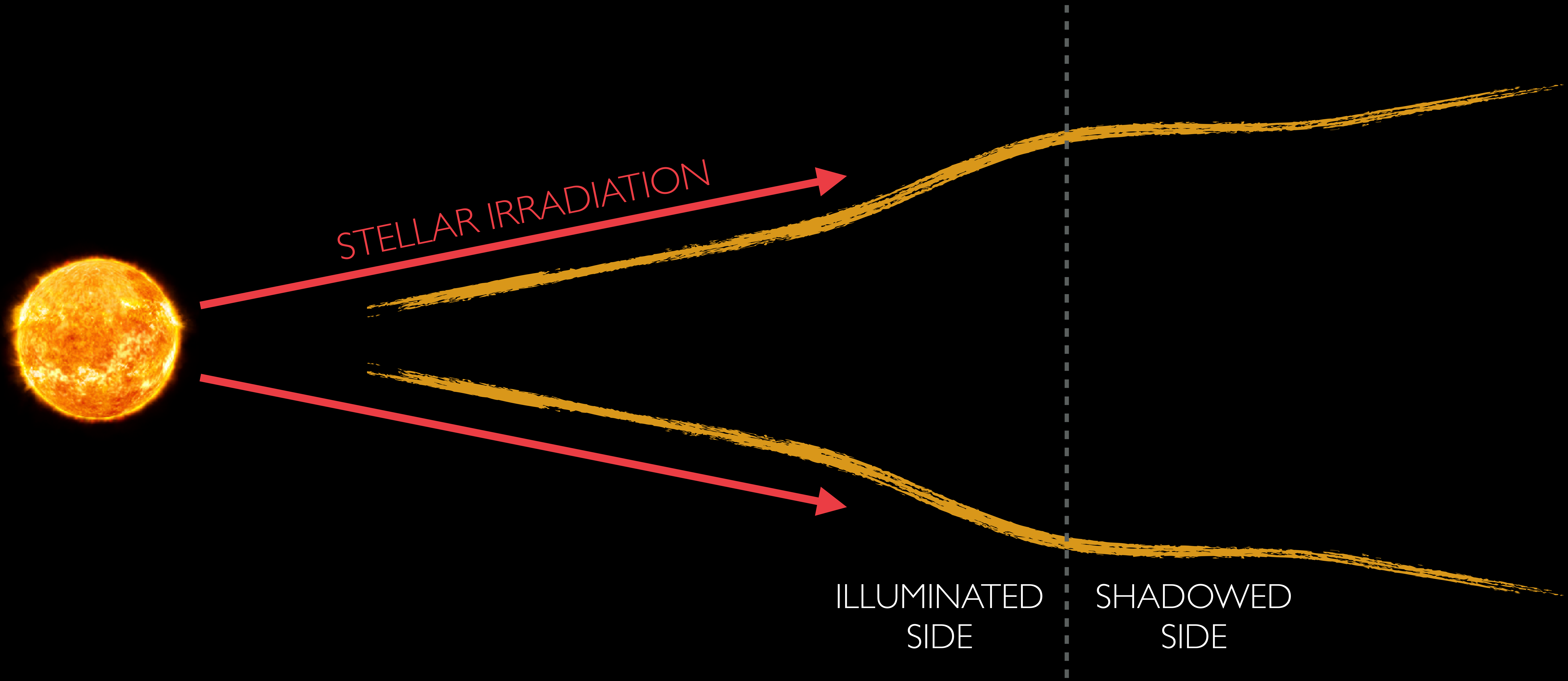

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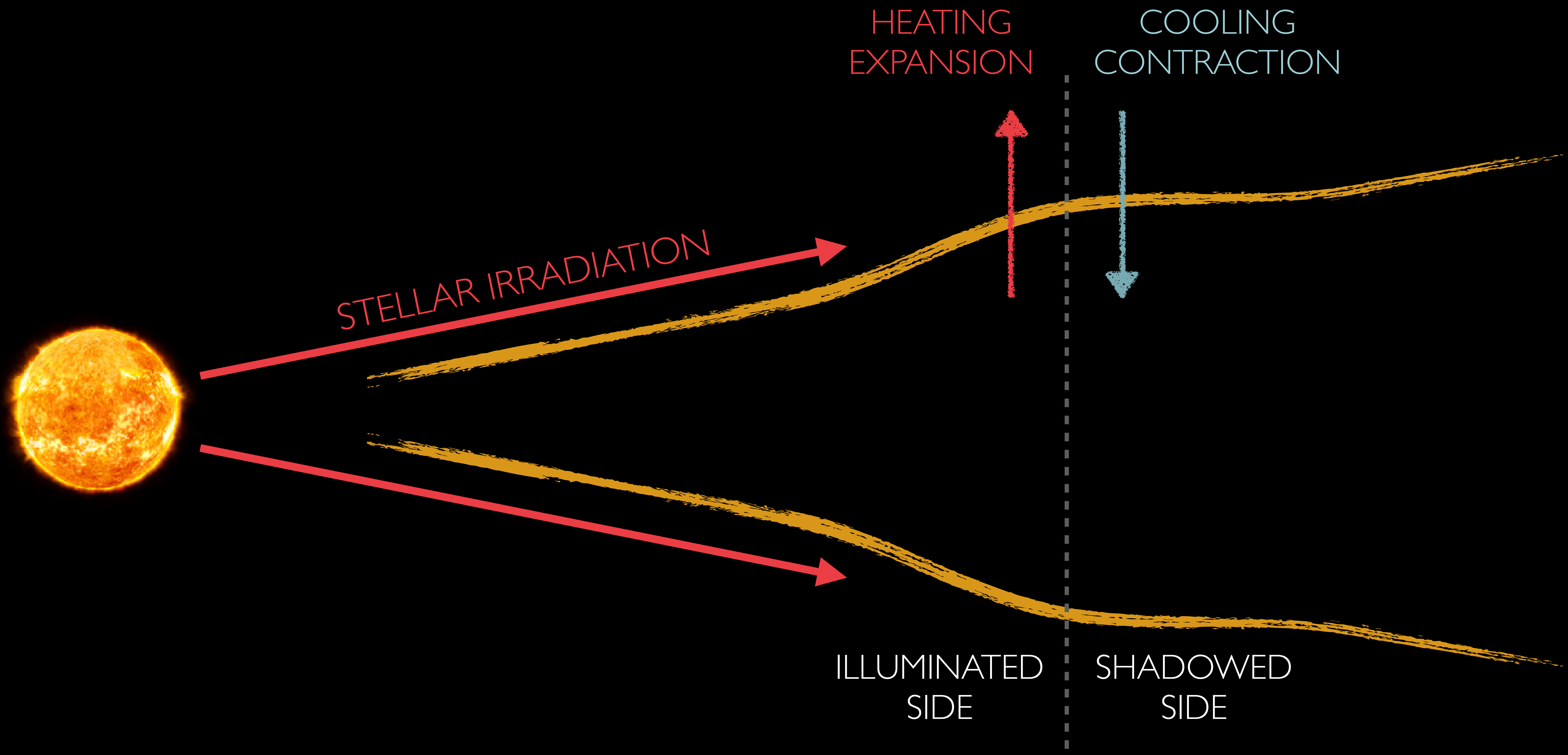
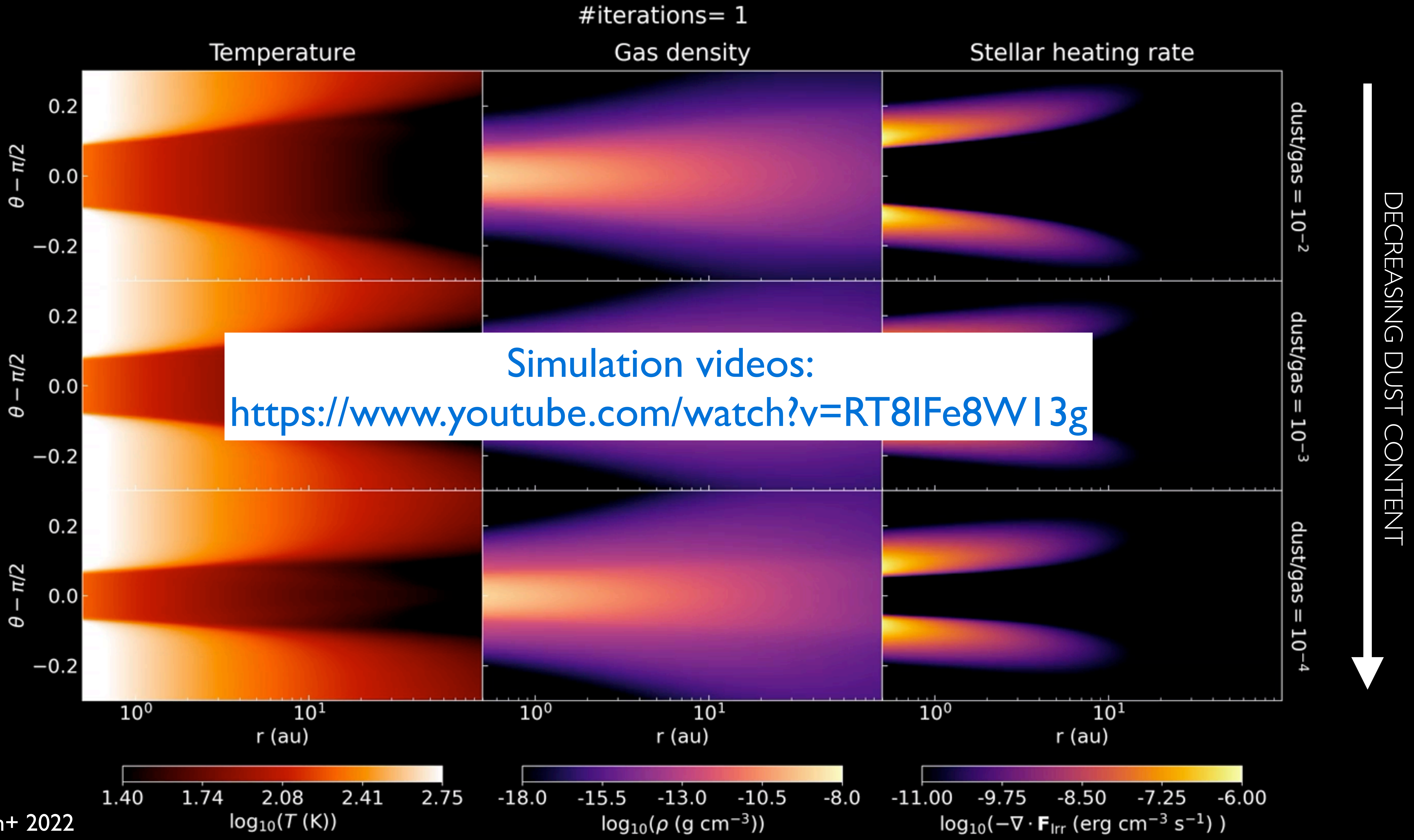


SELF-SHADOWING INSTABILITY?

INITIAL CONDITIONS - HYDROSTATIC EQUILIBRIUM



Simulation videos:
https://www.youtube.com/watch?v=RT8IFe8W13g

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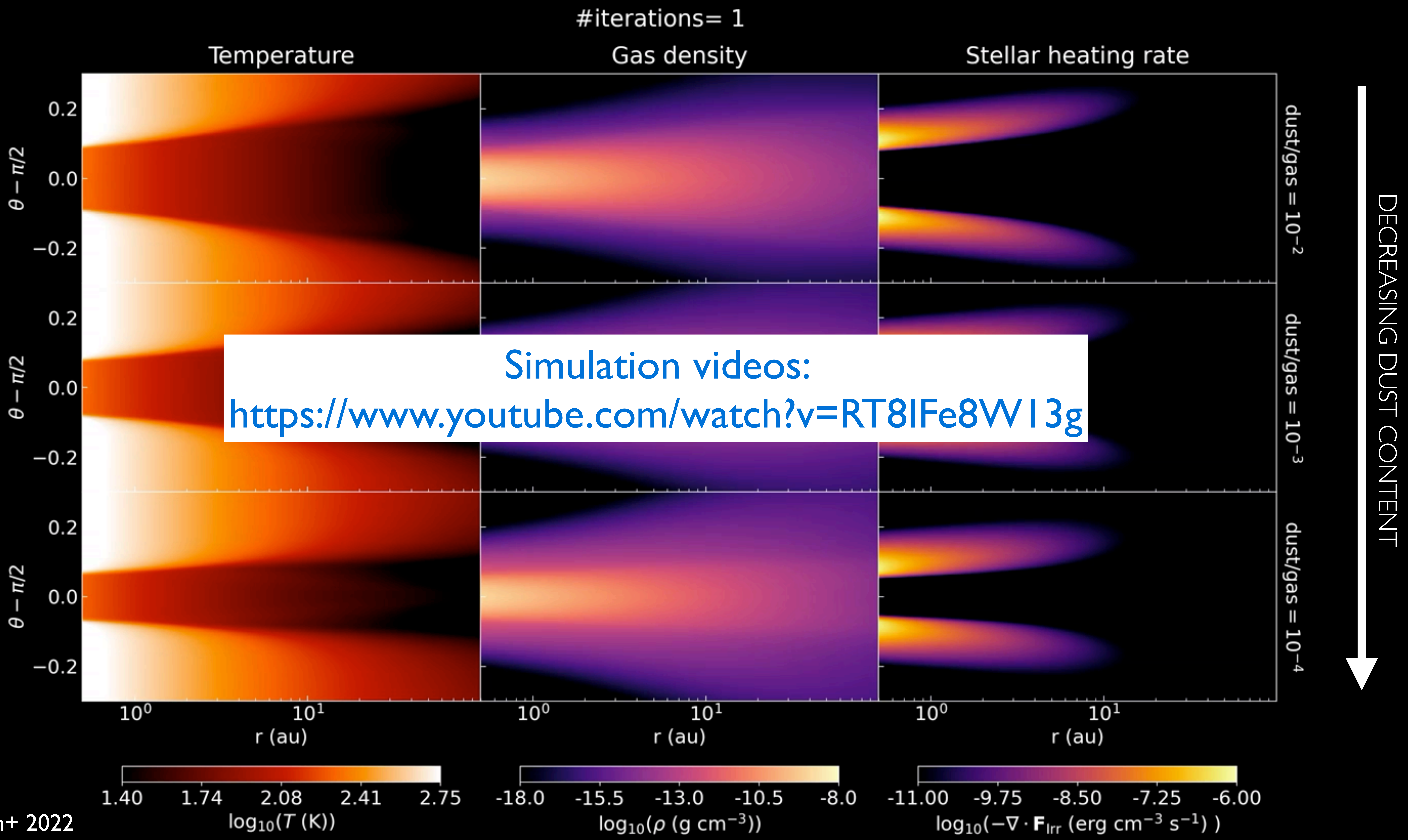

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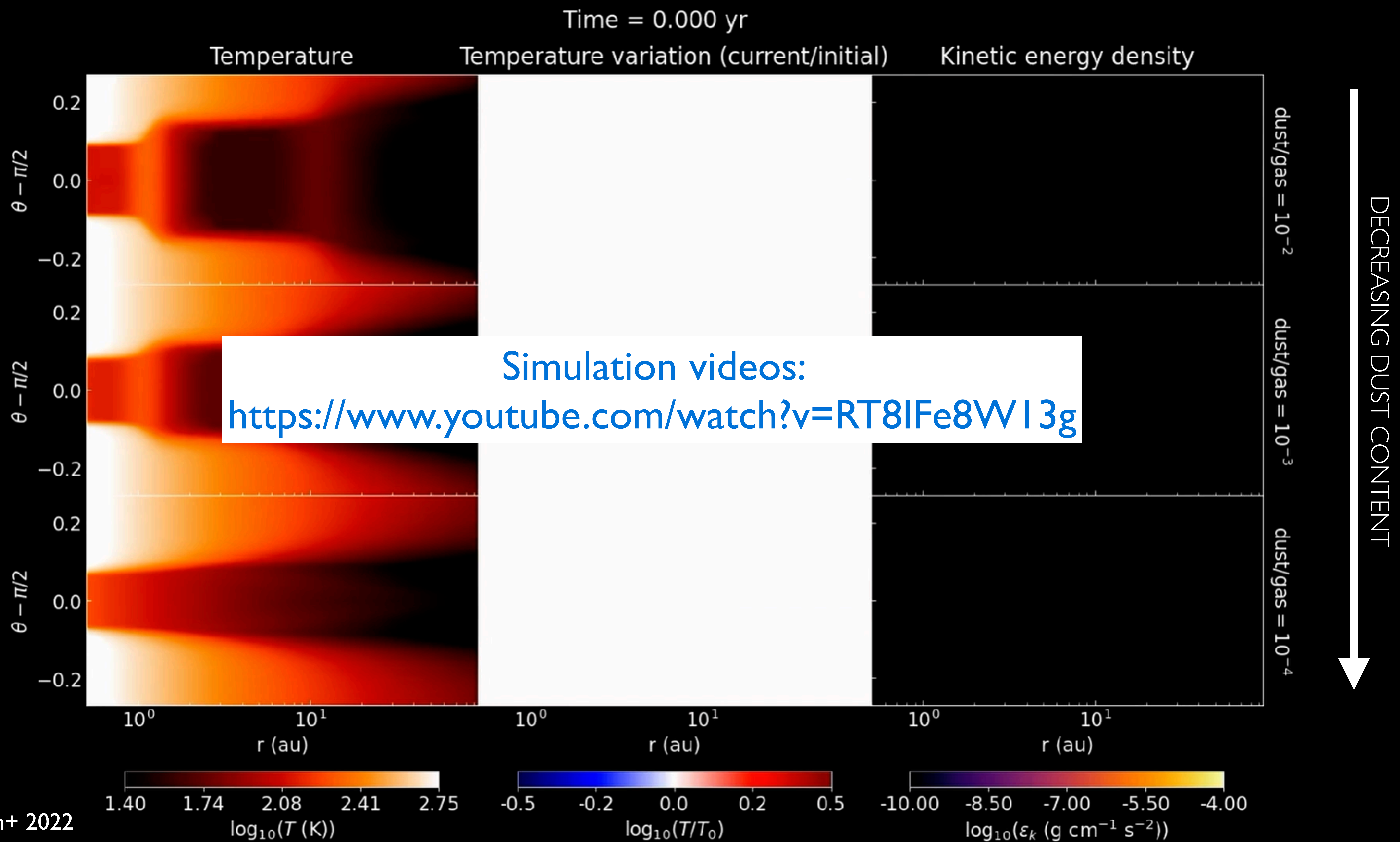
RADIATION HYDRODYNAMICS



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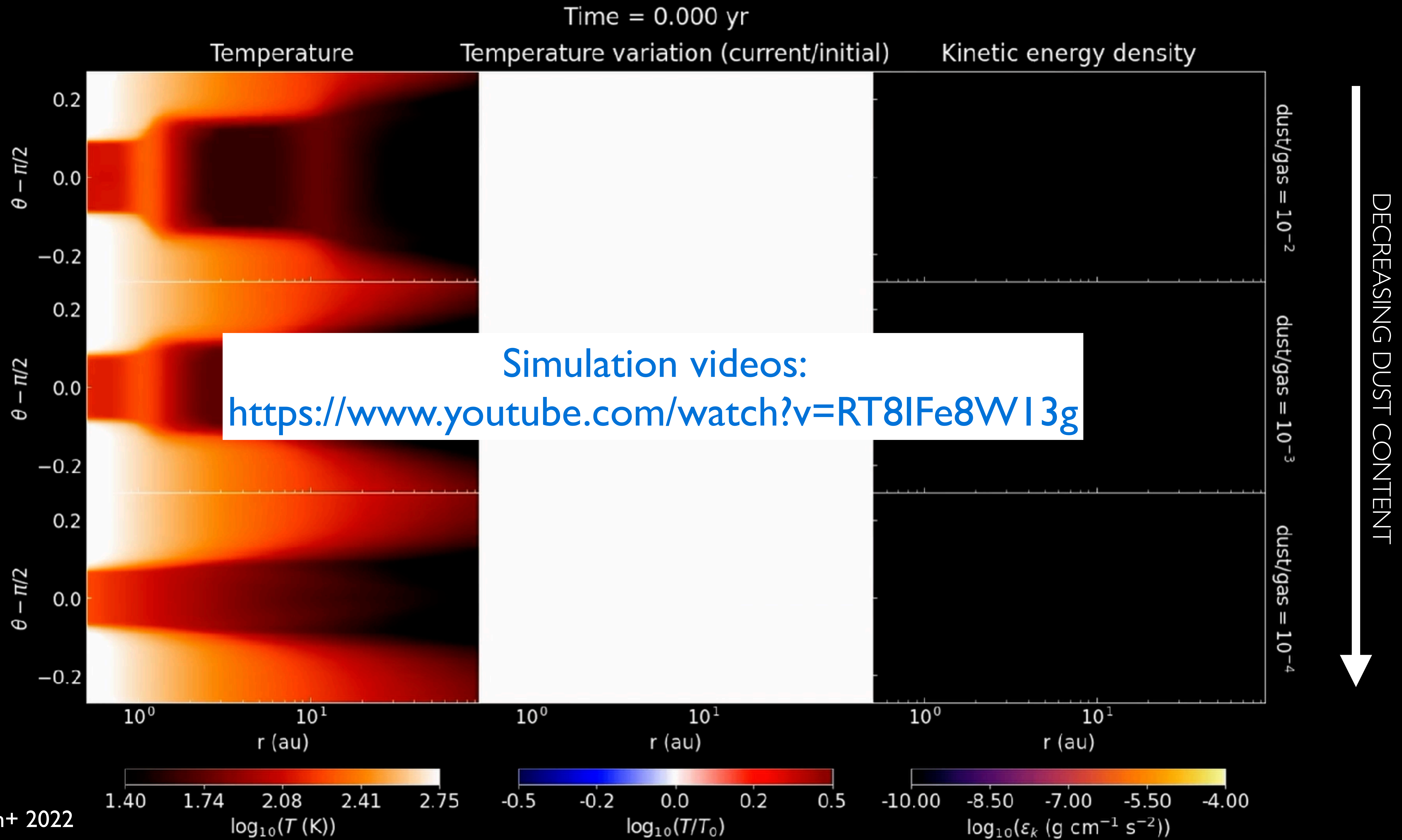

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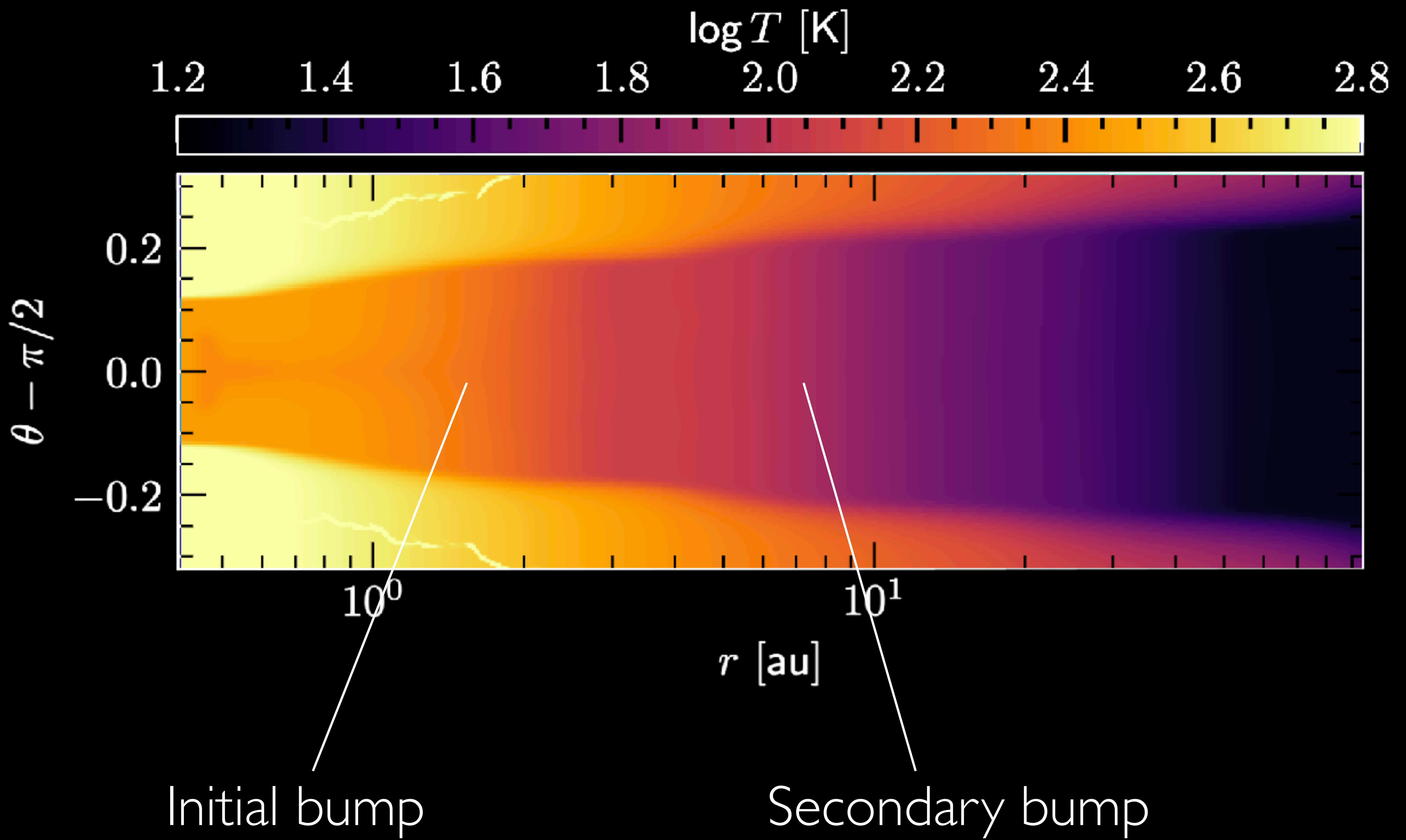
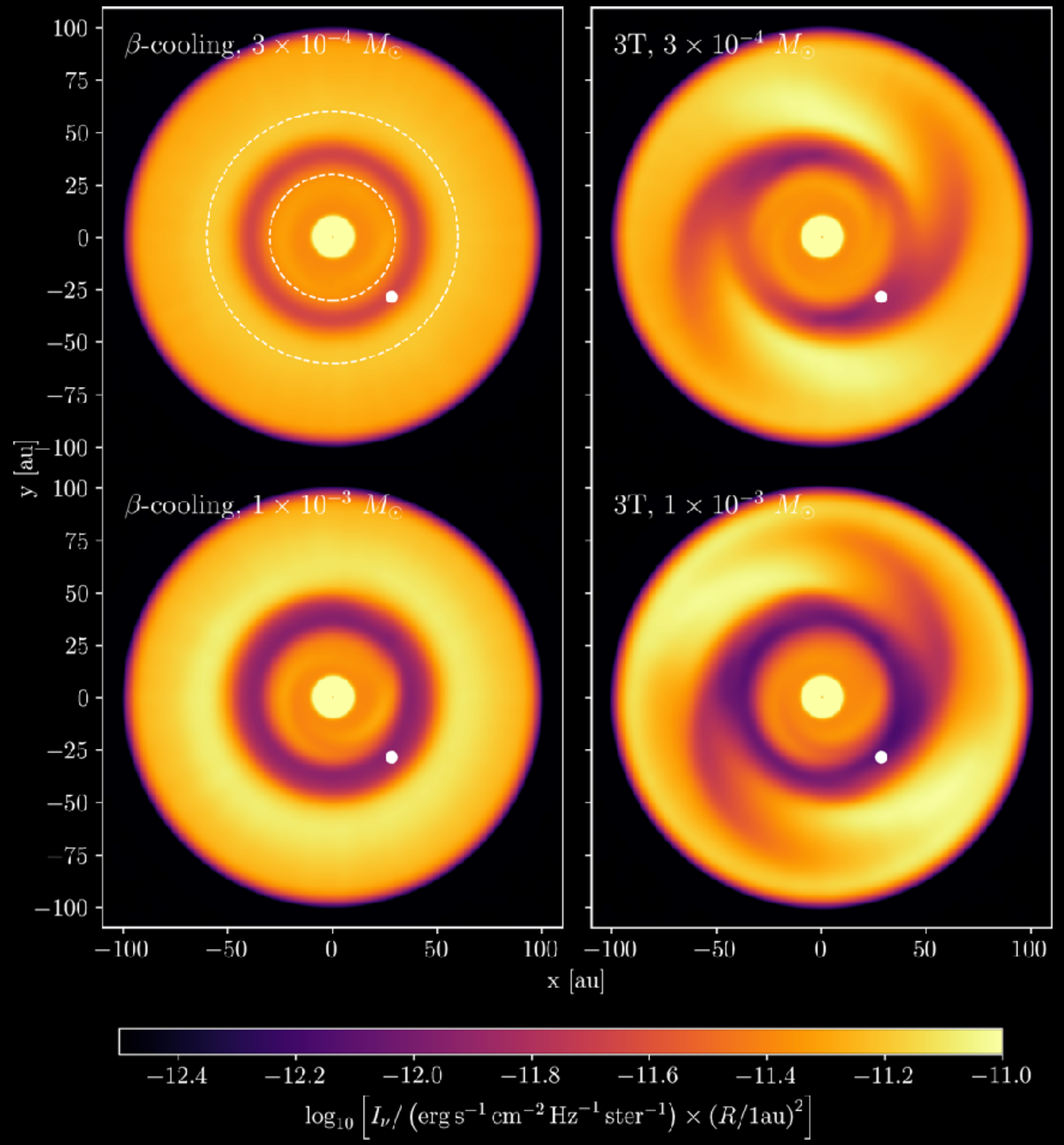
Bumps caused e.g. by opacity transitions/ice lines/sublimation fronts can produce a train of secondary bumps and shadows **(Sudarshan+ 2026)**:



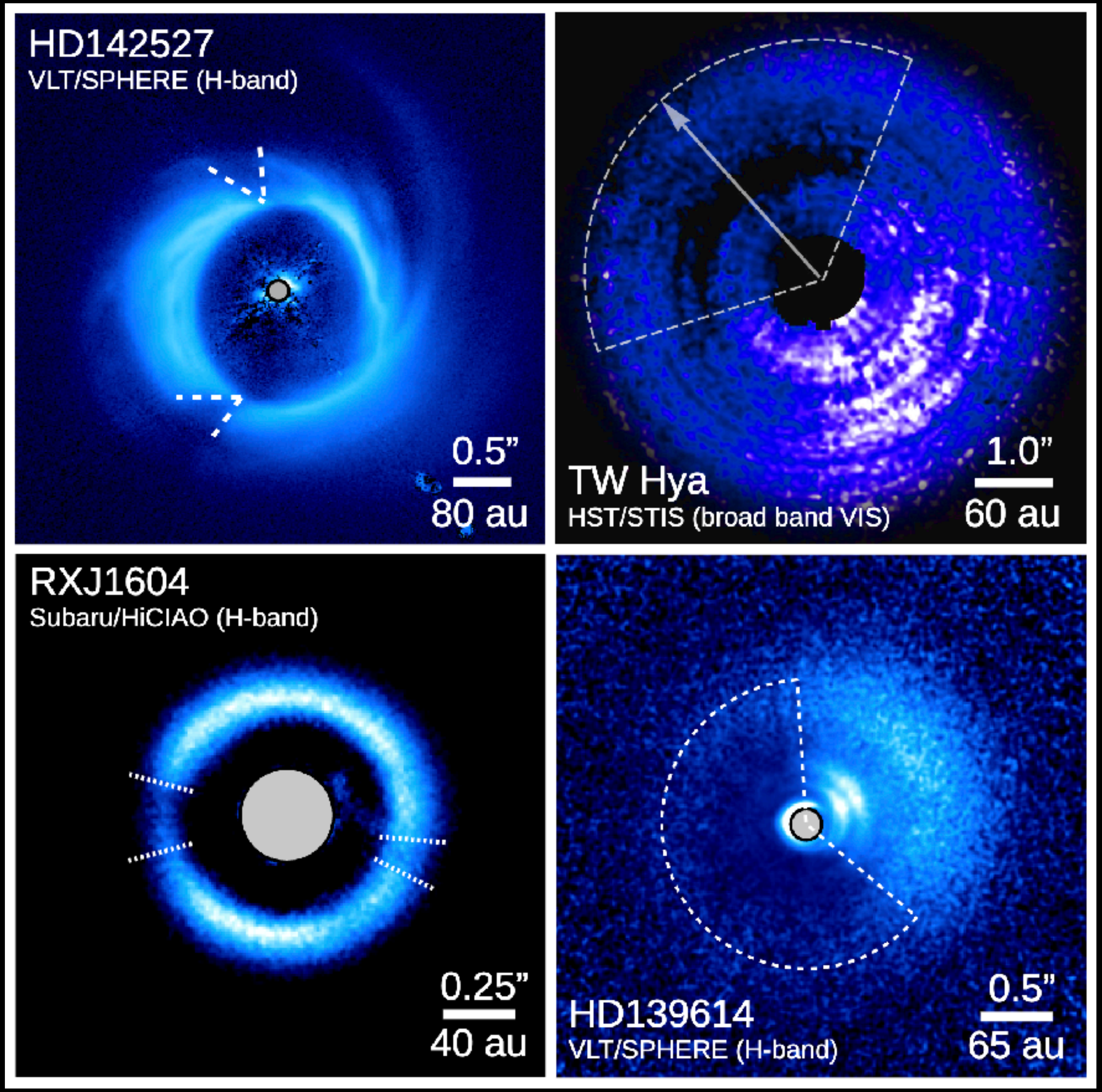
Non-axisymmetric structures can be amplified by stellar irradiation **(Muley+ 2024, 2026)**:

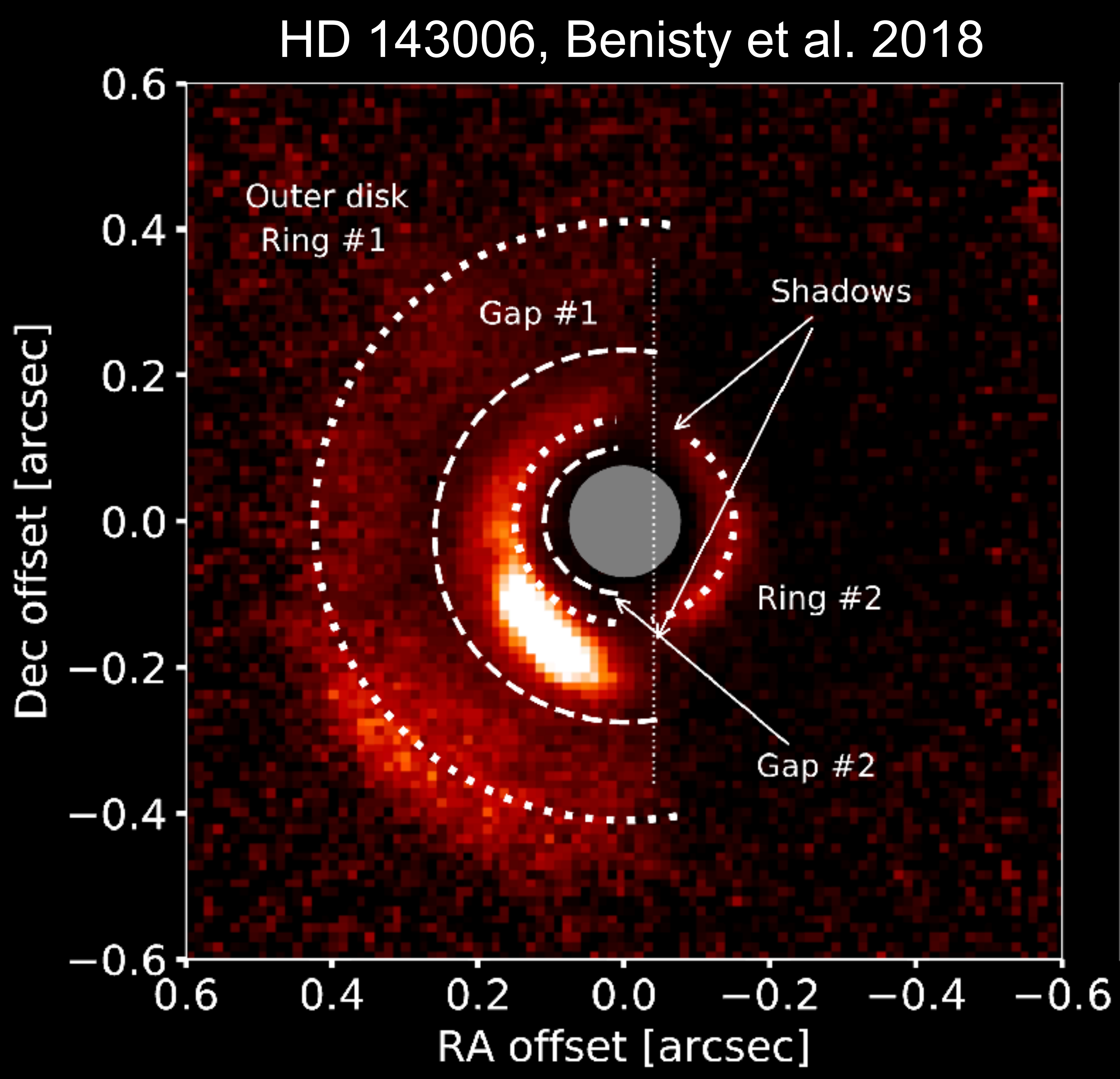
PRESCRIBED β-COOLING

STELLAR IRRADIATION



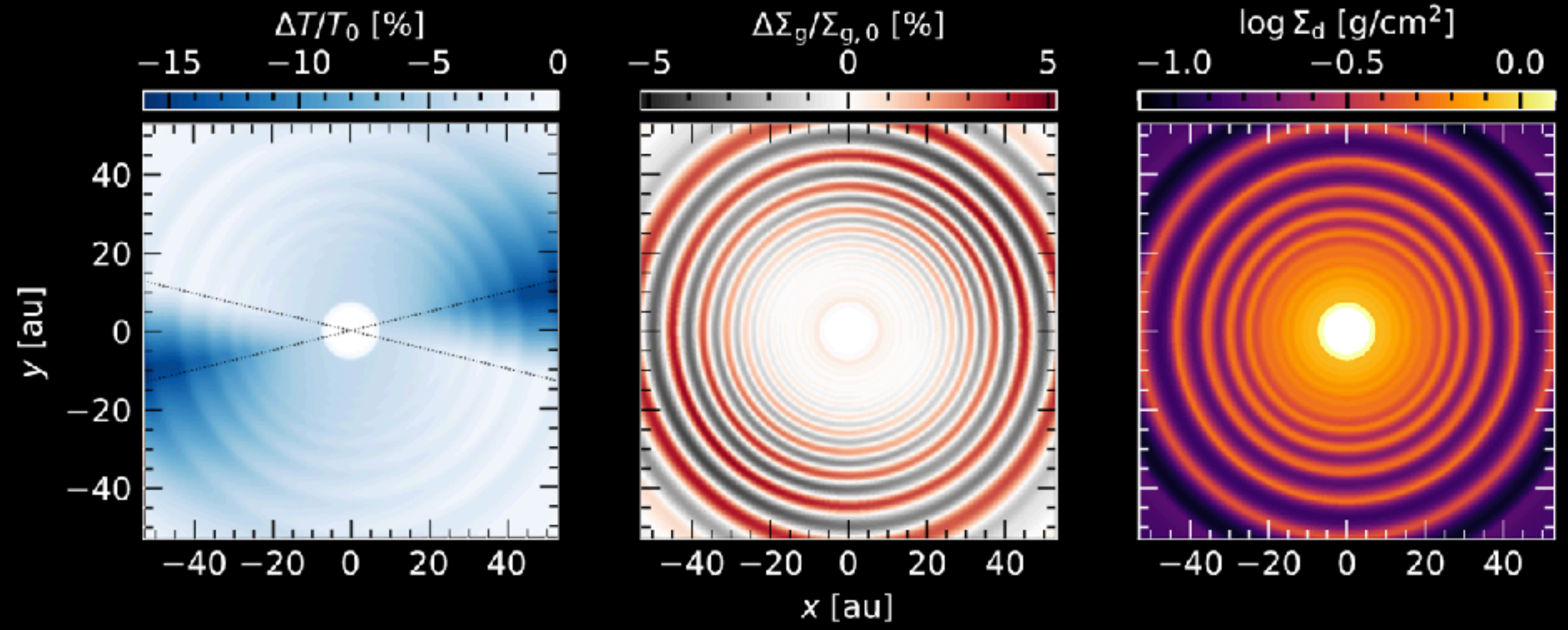
Shadows in protoplanetary disks



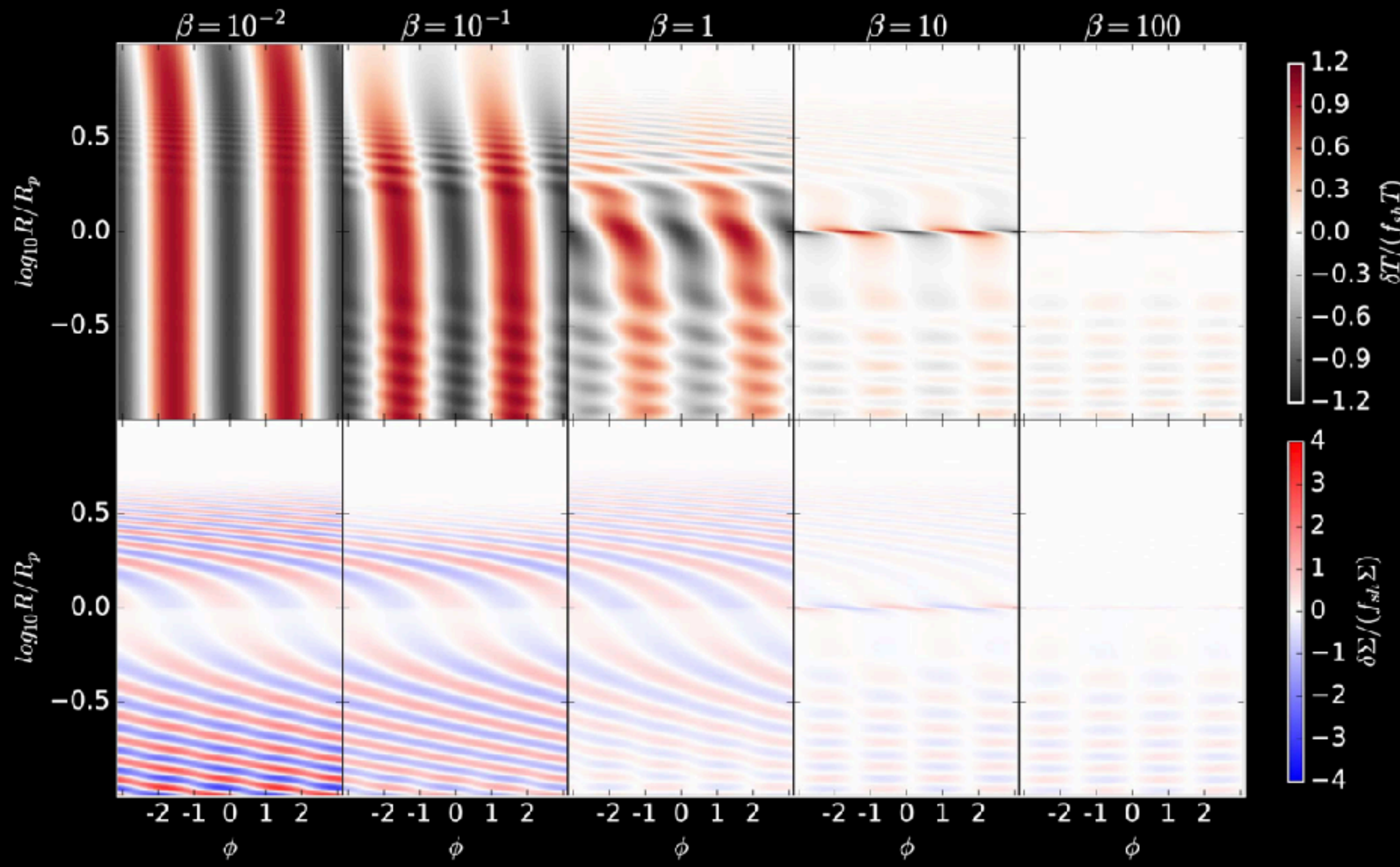


HD 142527 (Hunziker et al. 2021), RX J1604 (Mayama et al. 2012), TW Hya (Debes et al. 2017, image credit: ESA) and HD 139614 (Muro-Arena et al. 2020), from Benisty et al. 2023 (PPVII)

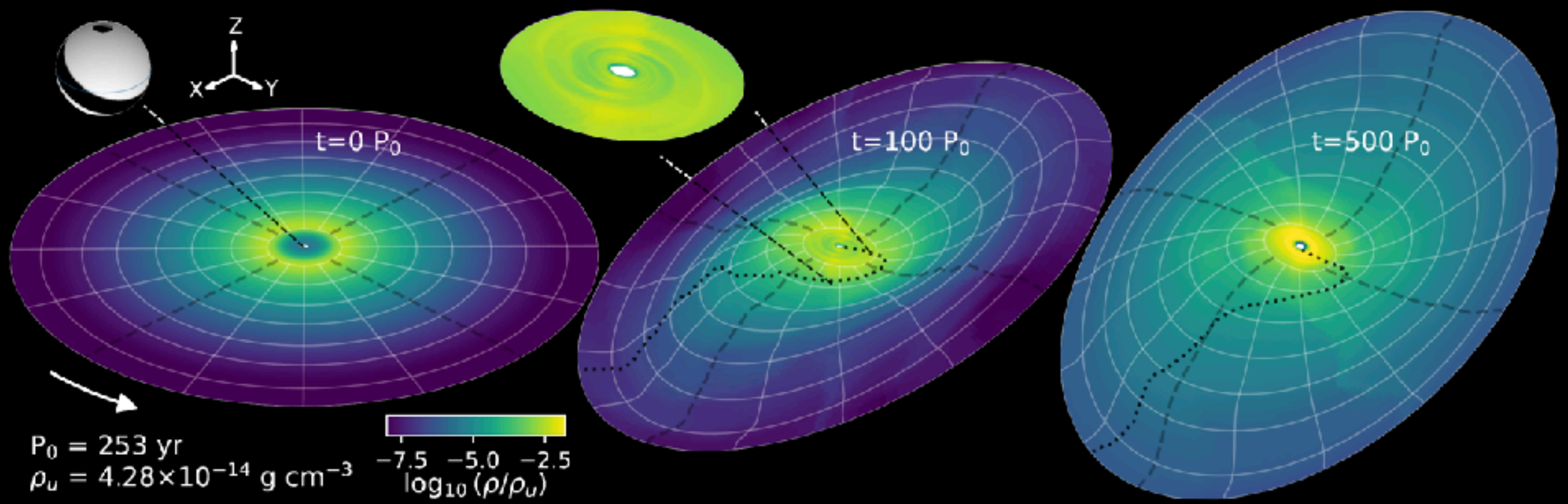
Non-axisymmetric shadows (e.g. by inner regions) should have dynamical consequences:



Rings (Ziampras+ 2025)



Spirals (Zhu+ 2025, Zhang+ 2024)



Warps (Zhang+ 2025, last Tuesday's talk)

- Hydrostatic models modify dynamical timescales and can trigger instabilities themselves, but a self-shadowing instability is unlikely.

- Shadowing can still enhance/produce structures (e.g. non-axisymmetric) if perturbations are continuously enforced (planets, shadows from inner regions, other instabilities like Rossby Wave Instability).

VERTICAL-SHEAR INSTABILITY ANGULAR MOMENTUM REDISTRIBUTION

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VERTICAL-SHEAR INSTABILITY
ANGULAR MOMENTUM REDISTRIBUTION

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VERTICAL-SHEAR INSTABILITY
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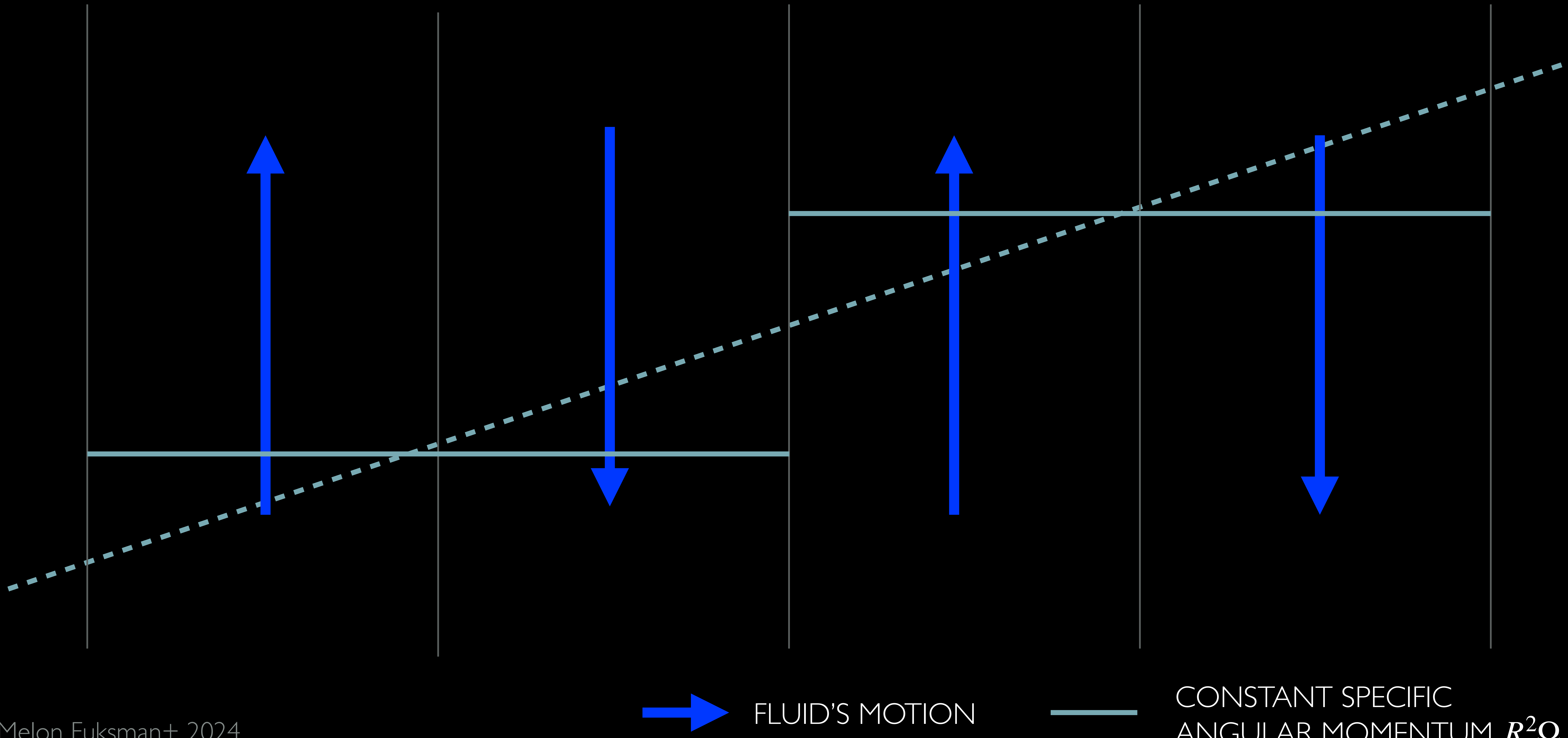

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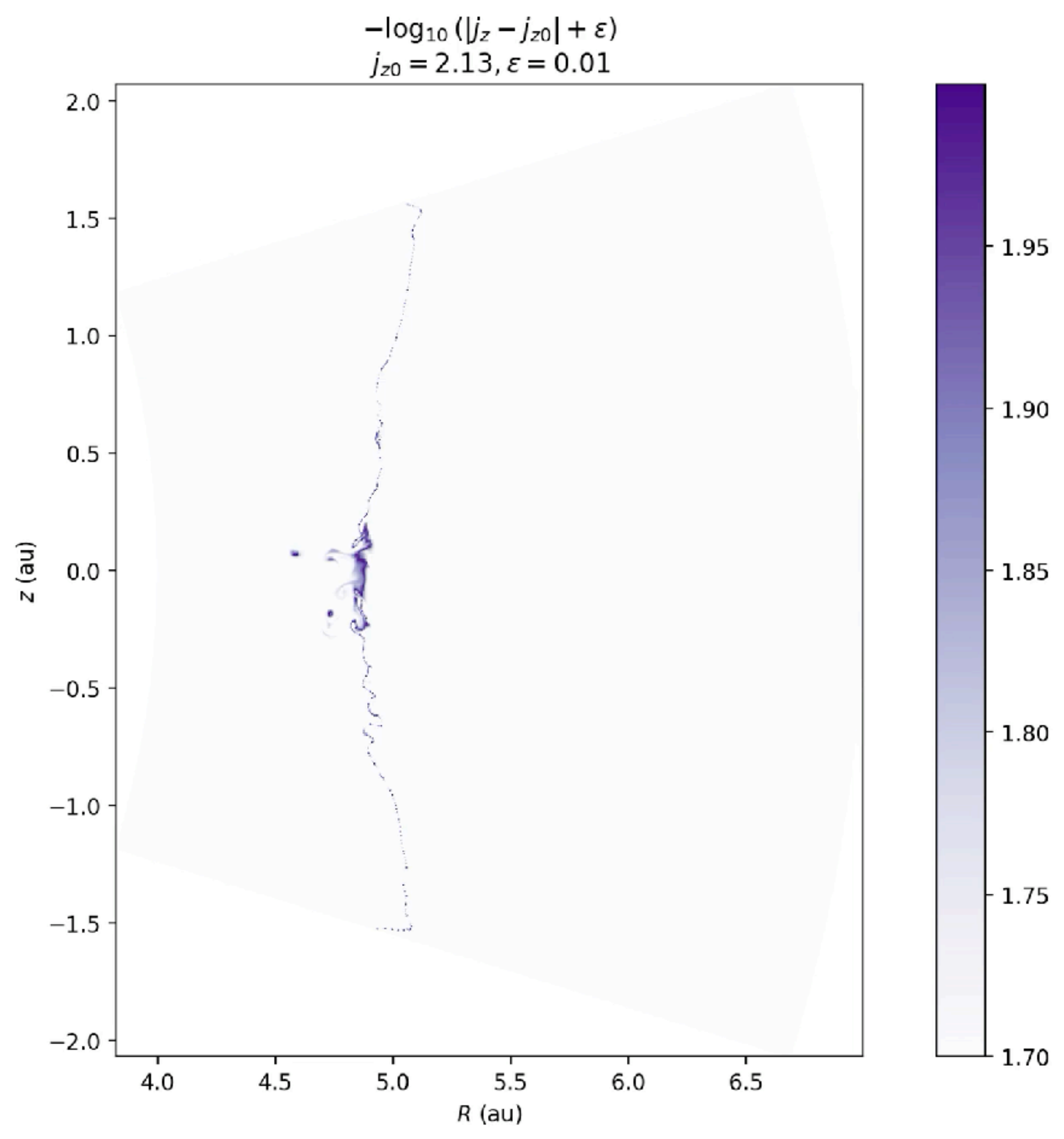
ANGULAR MOMENTUM DIFFUSION

The specific angular momentum, $j_z = Rv_\phi$, should be conserved along streamlines for zero numerical diffusion:

$$\frac{\partial j_z}{\partial t} + \mathbf{v} \cdot \nabla j_z = \nabla \cdot \left(\nu_{\text{num}} \nabla j_z\right)$$



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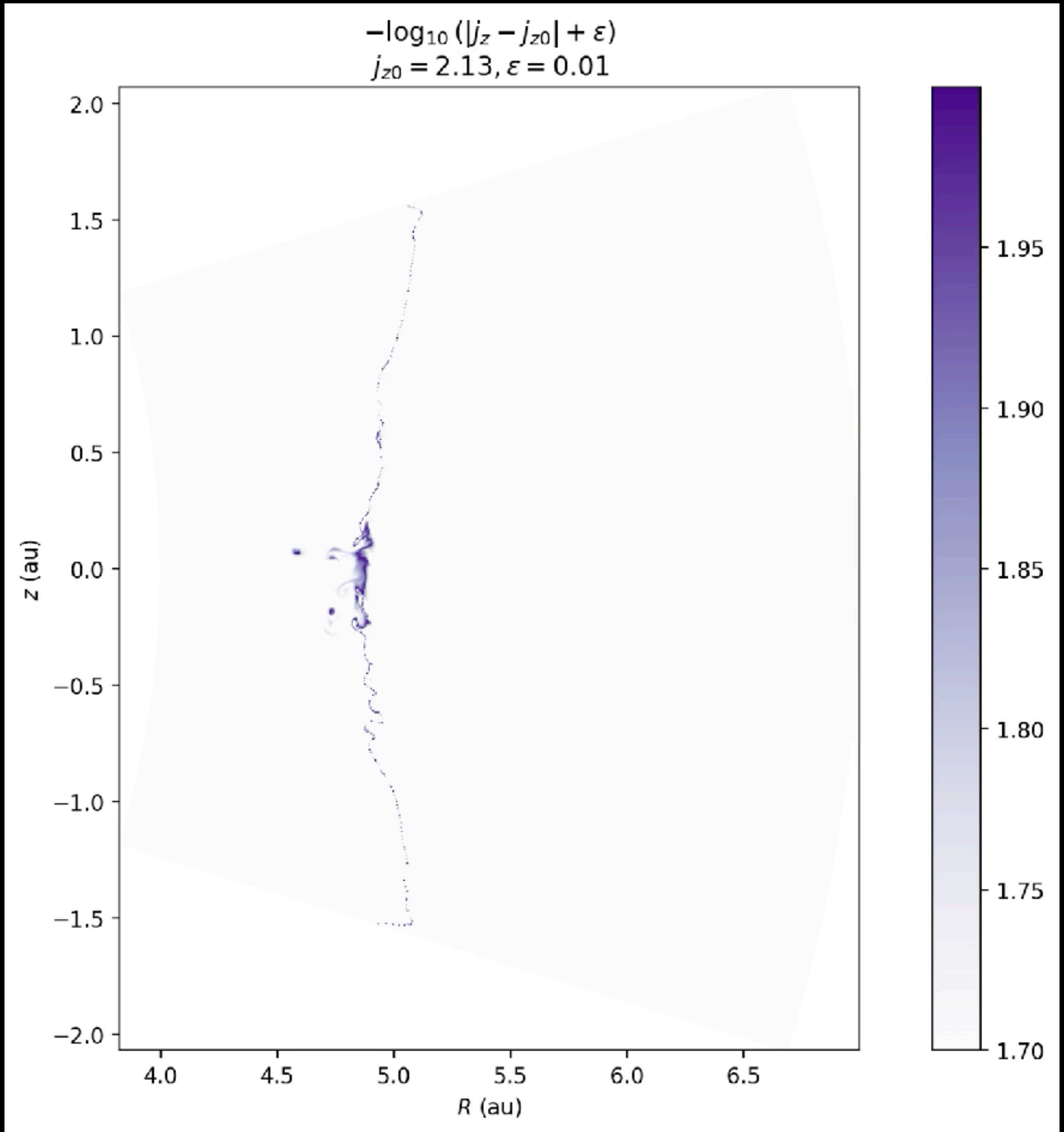
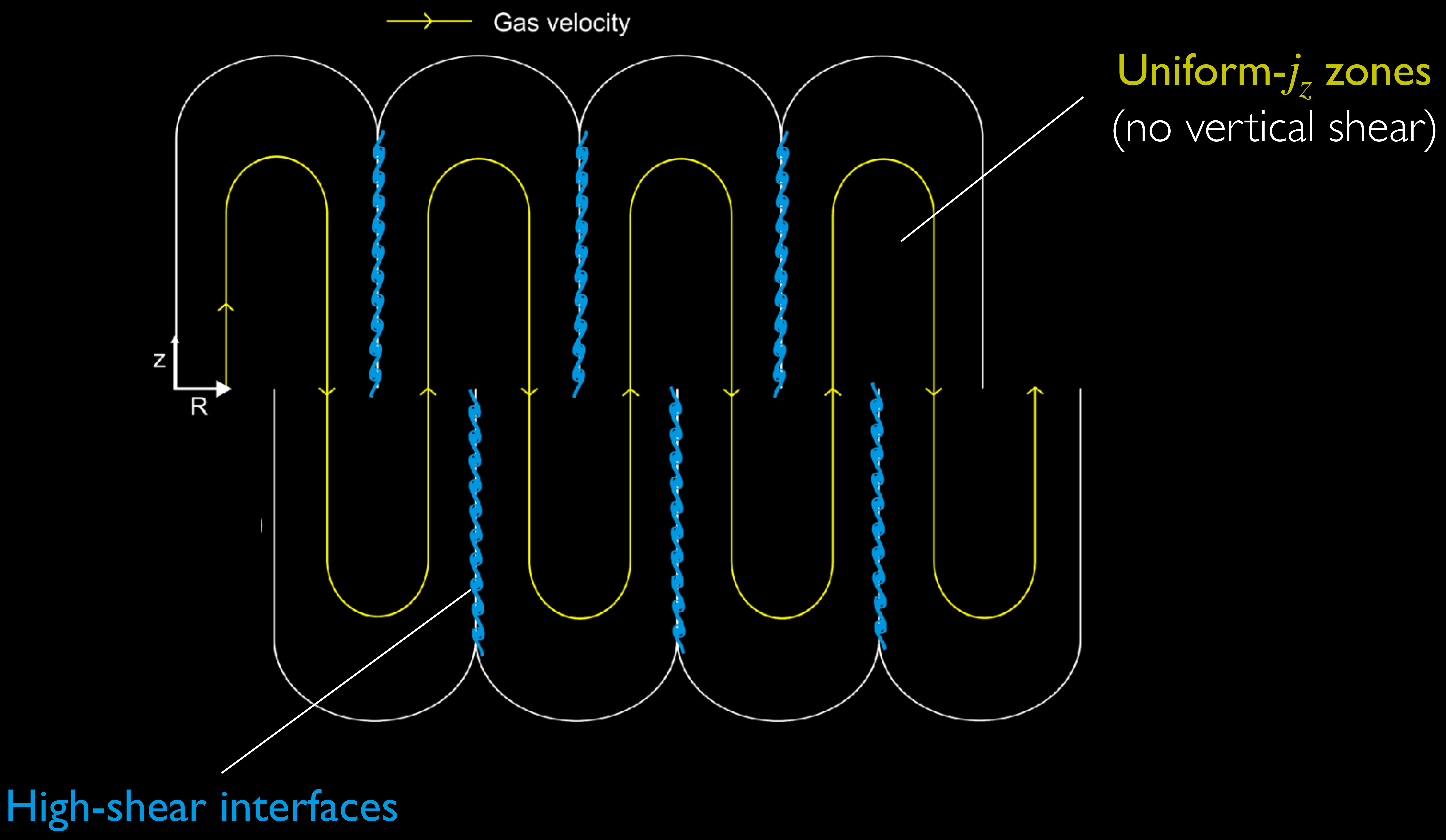
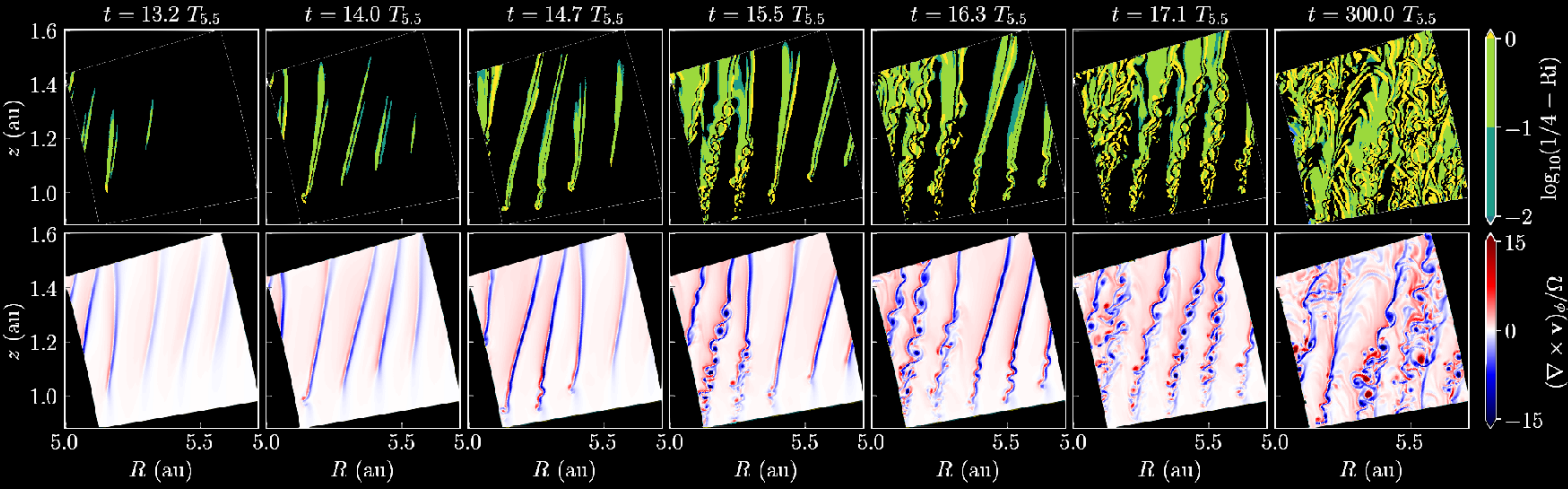



Gas velocity
z
R
Uniform-j_z zones
(no vertical shear)
High-shear interfaces

SECONDARY INSTABILITIES - KELVIN-HELMHOLTZ INST.

Time →



KHI condition

$\phi-$ vorticity

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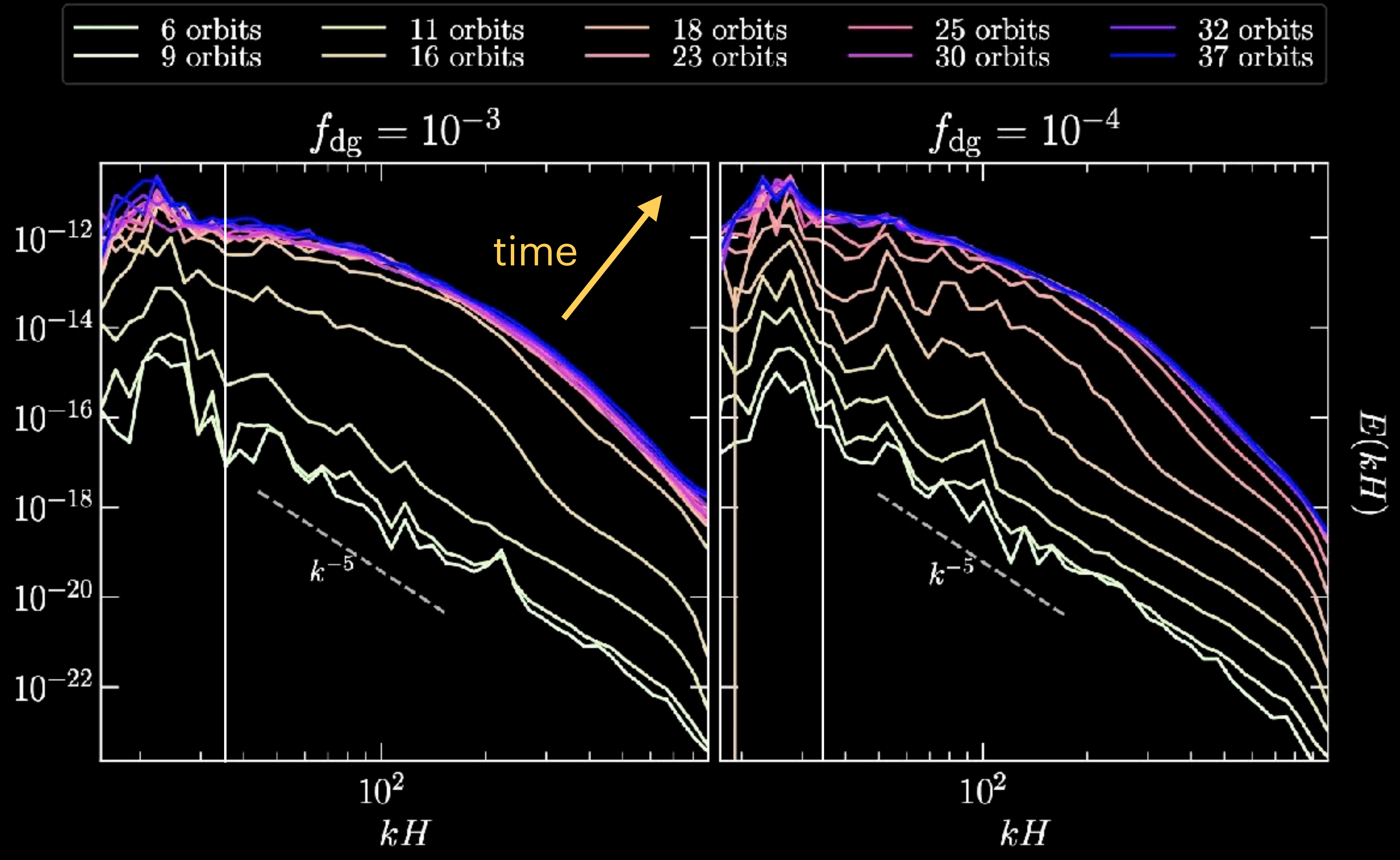

Linear growth stage: 95% of kinetic energy in large scales

Saturated stage: 50-60% of kinetic energy in large scales

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VORTICITY GENERATION

$$\frac{d}{dt}\int_S \underbrace{\omega_\phi}_{\text{Meridional vorticity}} dS = \int_S \left(\underbrace{\frac{\nabla\rho\times\nabla p}{\rho^2}}_{\text{Baroclinic torque}} + \underbrace{R\partial_z\Omega^2\hat{\phi}}_{\text{Centrifugal torque}} \right)\cdot d\mathbf{S}$$

VORTICITY GENERATION

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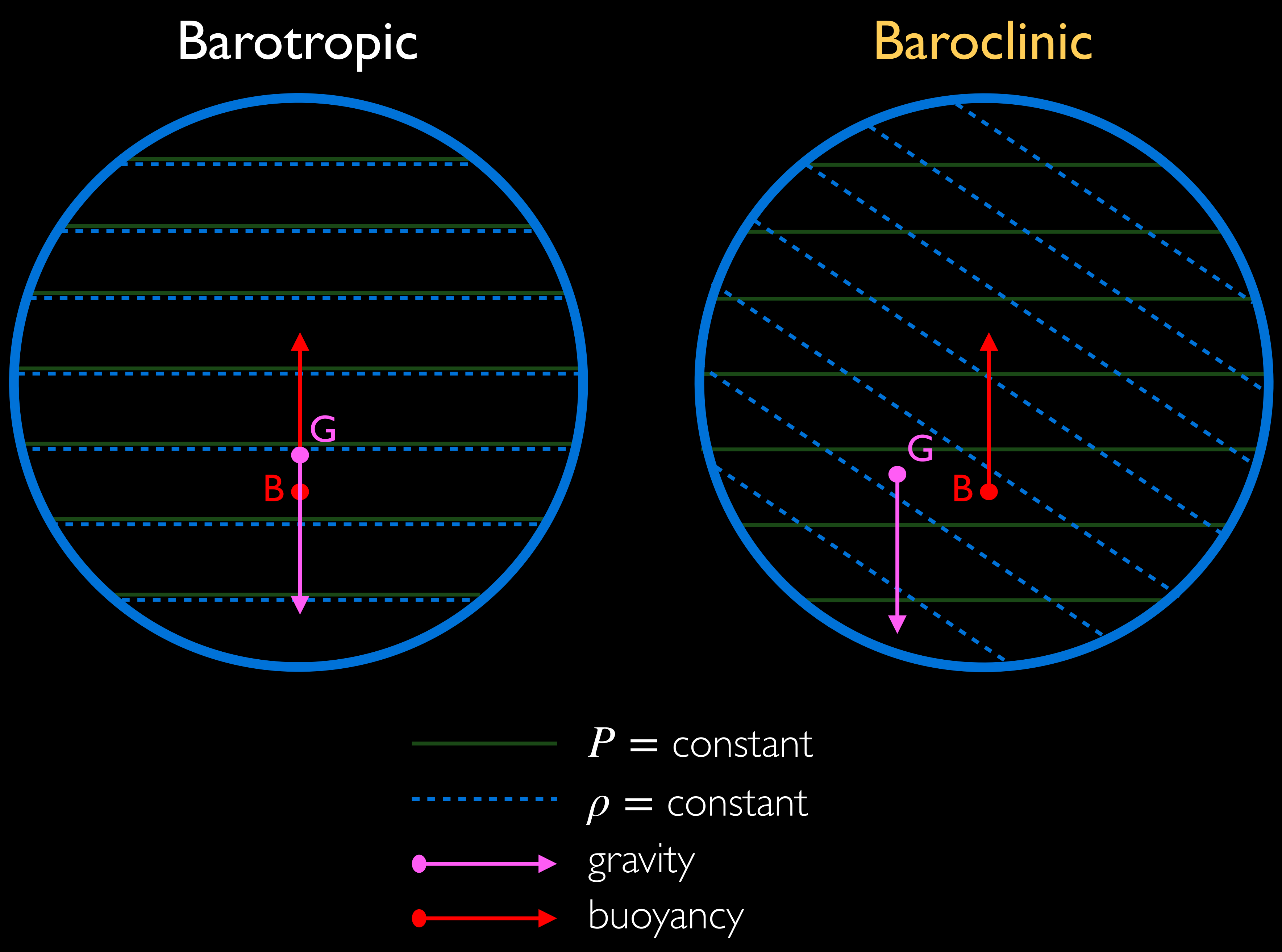
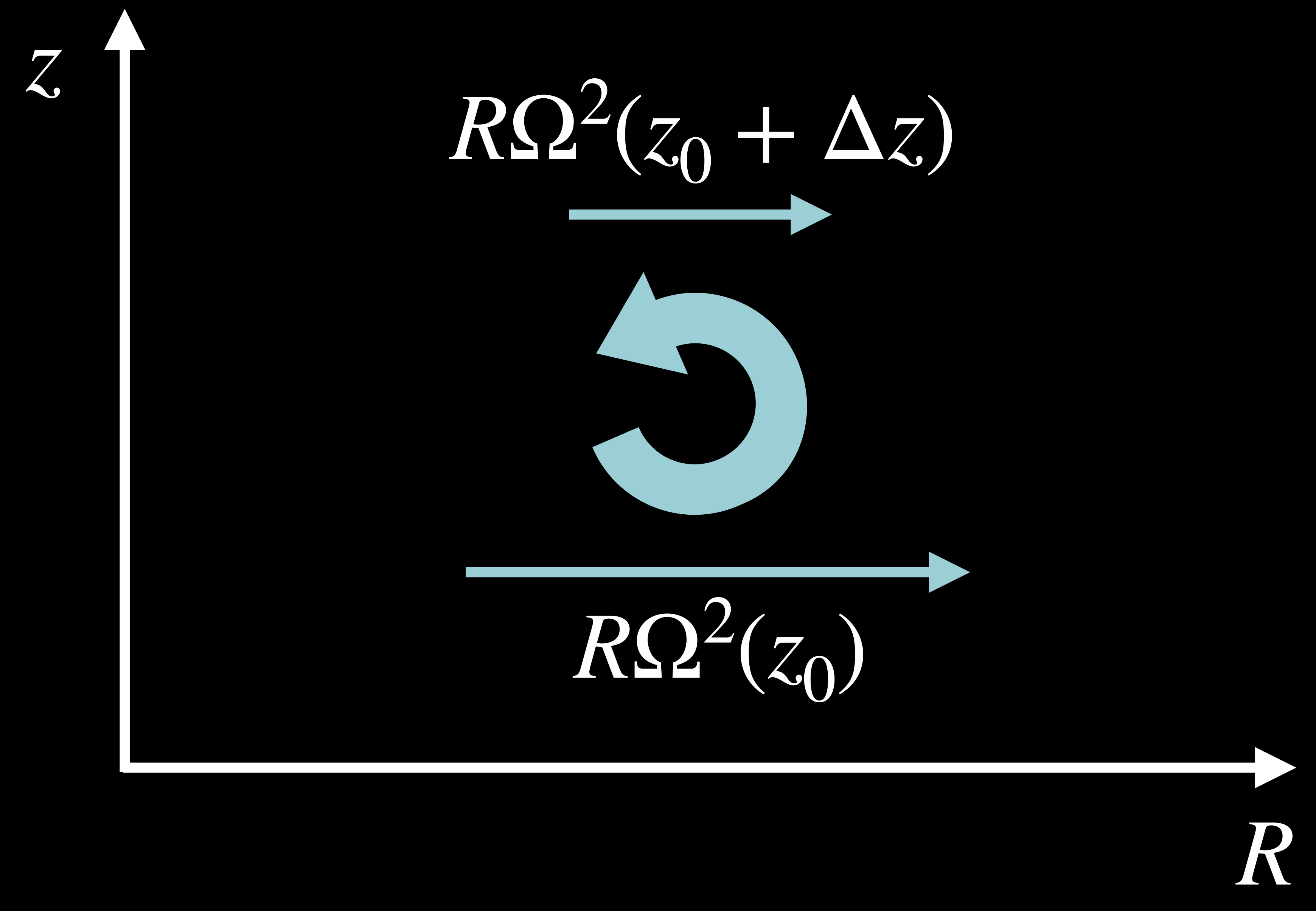
Baroclinic torque

$\nabla\rho\times\nabla p = 0$ $\qquad$ $\nabla\rho\times\nabla p \neq 0$



Centrifugal torque



VORTICITY GENERATION

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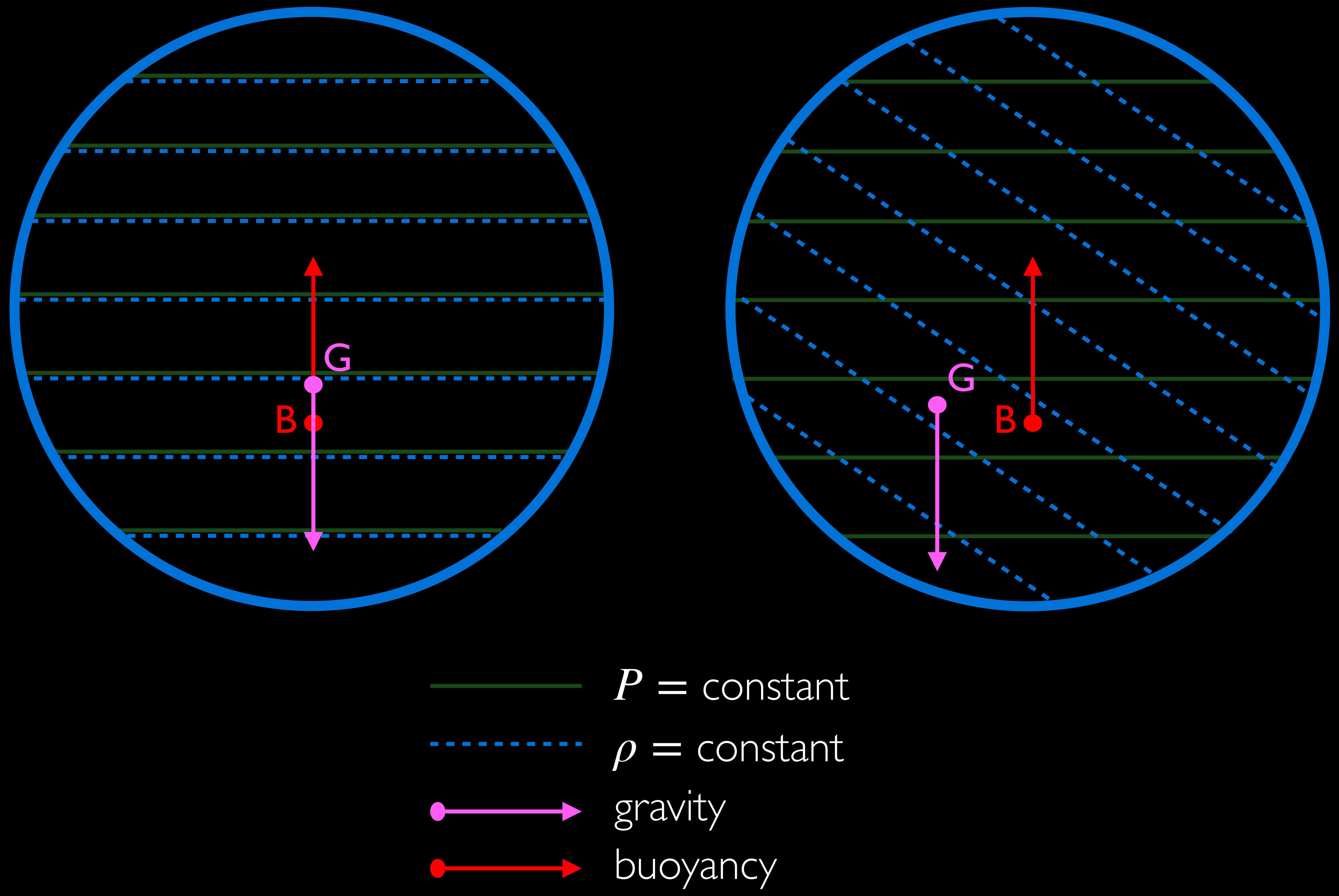
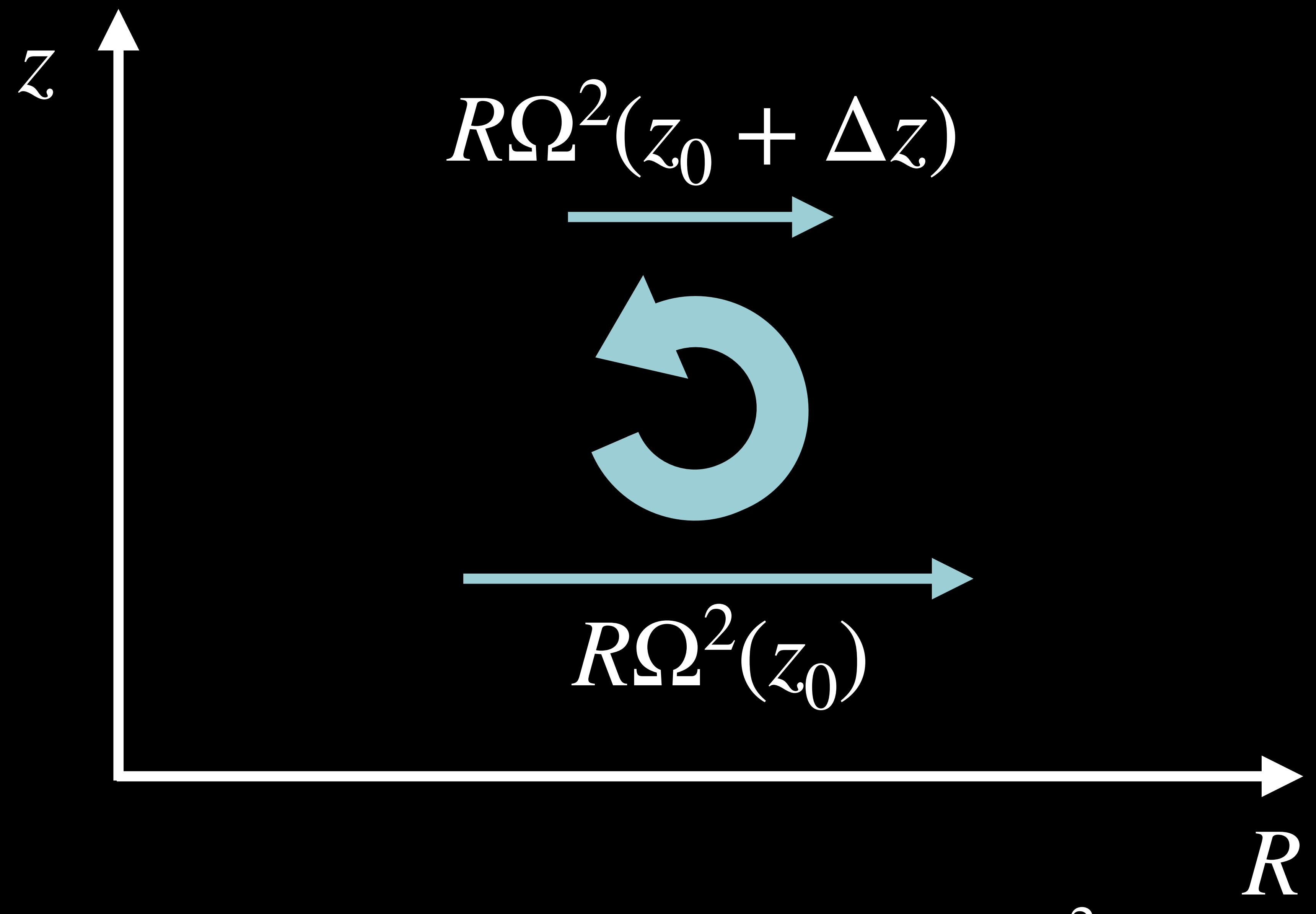
Baroclinic torque

$\nabla\rho \times \nabla p = 0$ $\nabla\rho \times \nabla p \neq 0$

Barotropic Baroclinic



Centrifugal torque



If uniform ang. mom. density $R^2\Omega$

$\rightarrow R\partial_z\Omega^2 = 0$

VORTICITY GENERATION

$$\frac{d}{dt}\int_S \underline{\omega_\phi}\, dS = \int_S \left(\underline{\frac{\nabla\rho \times \nabla p}{\rho^2}} \right) \cdot d\mathbf{S}$$

Meridional vorticity

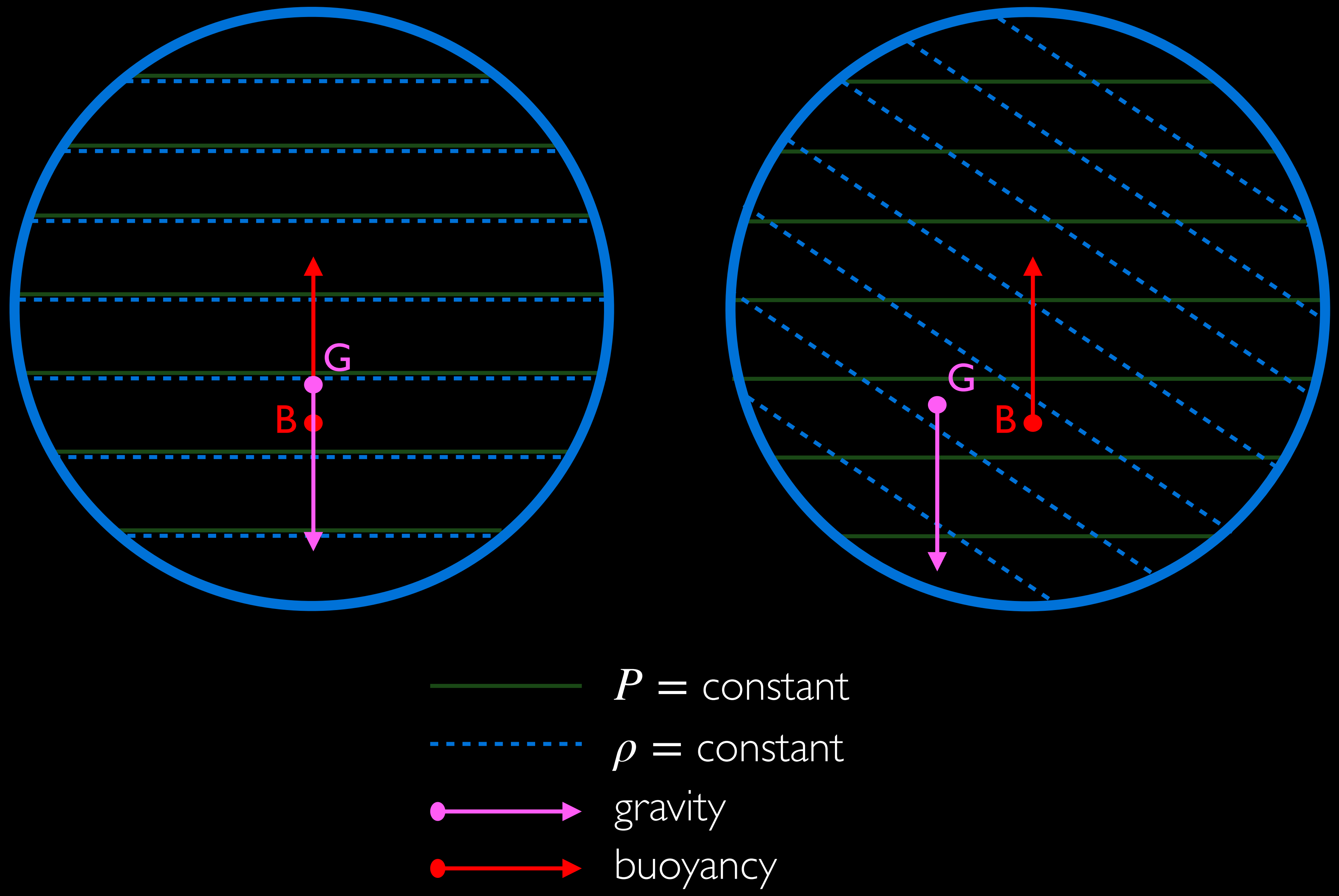
Baroclinic torque

Baroclinic torque

$\nabla\rho \times \nabla p = 0$ $\nabla\rho \times \nabla p \neq 0$

Barotropic

Baroclinic



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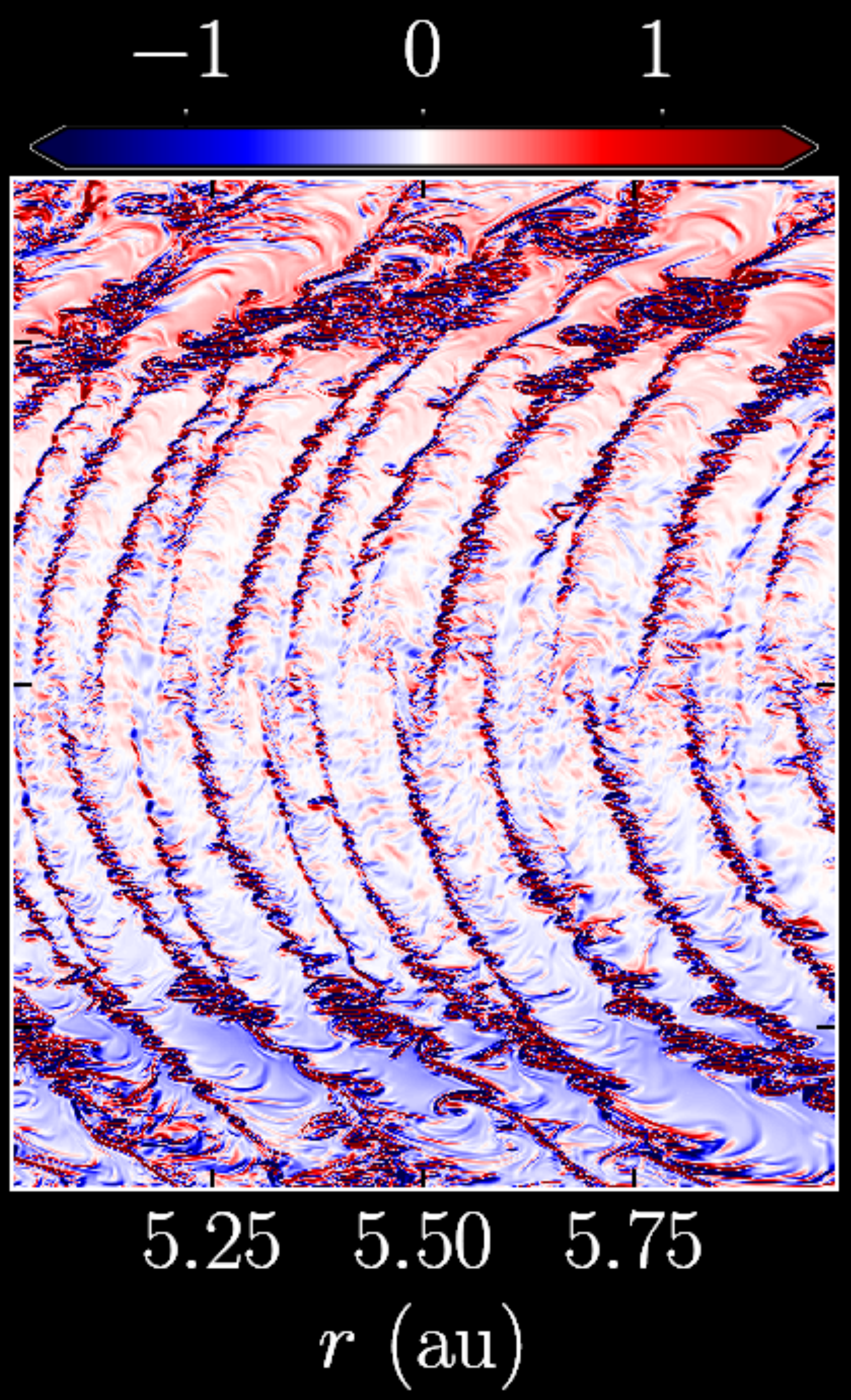
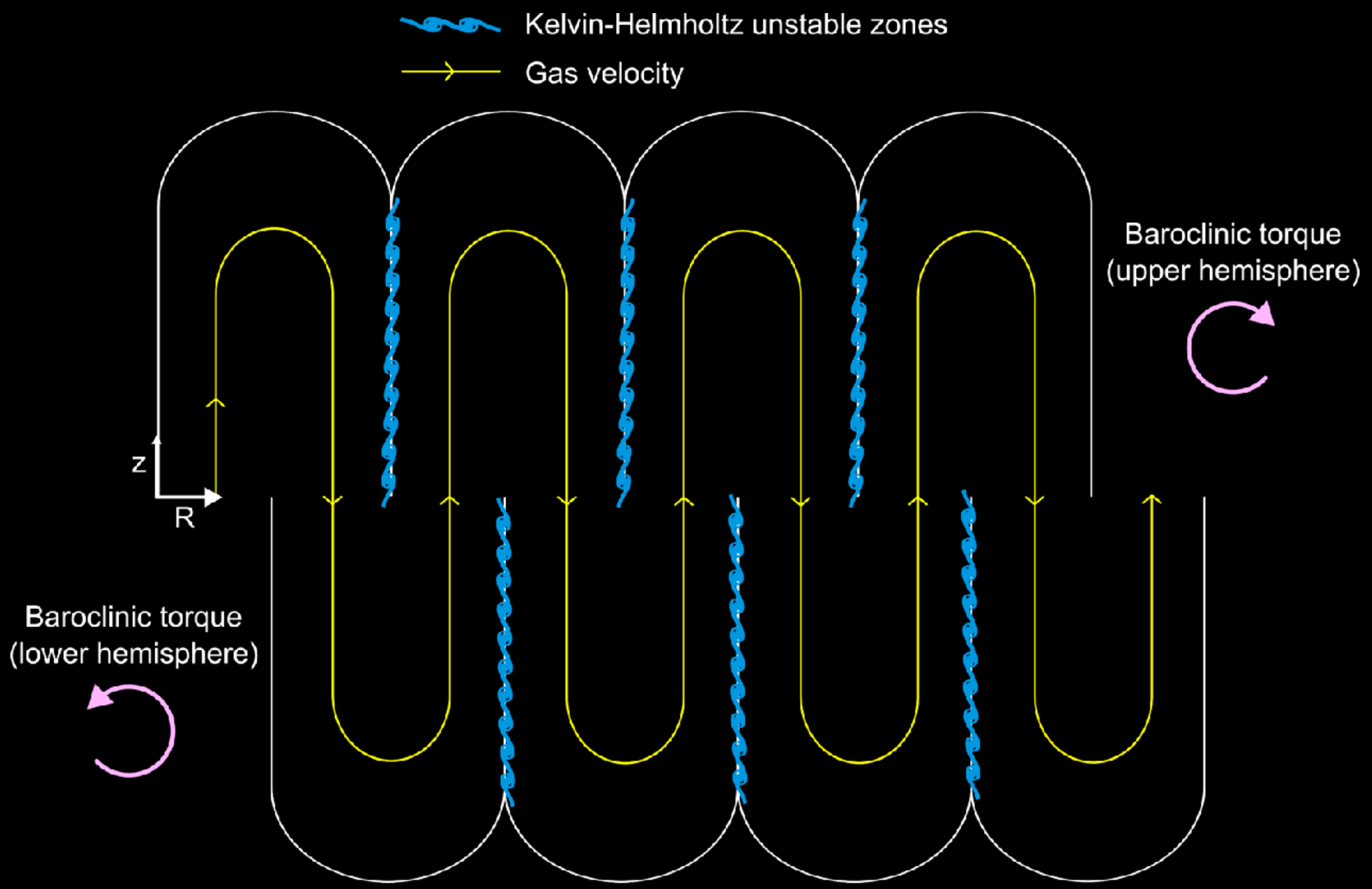
Baroclinic+centrifugal torque terms





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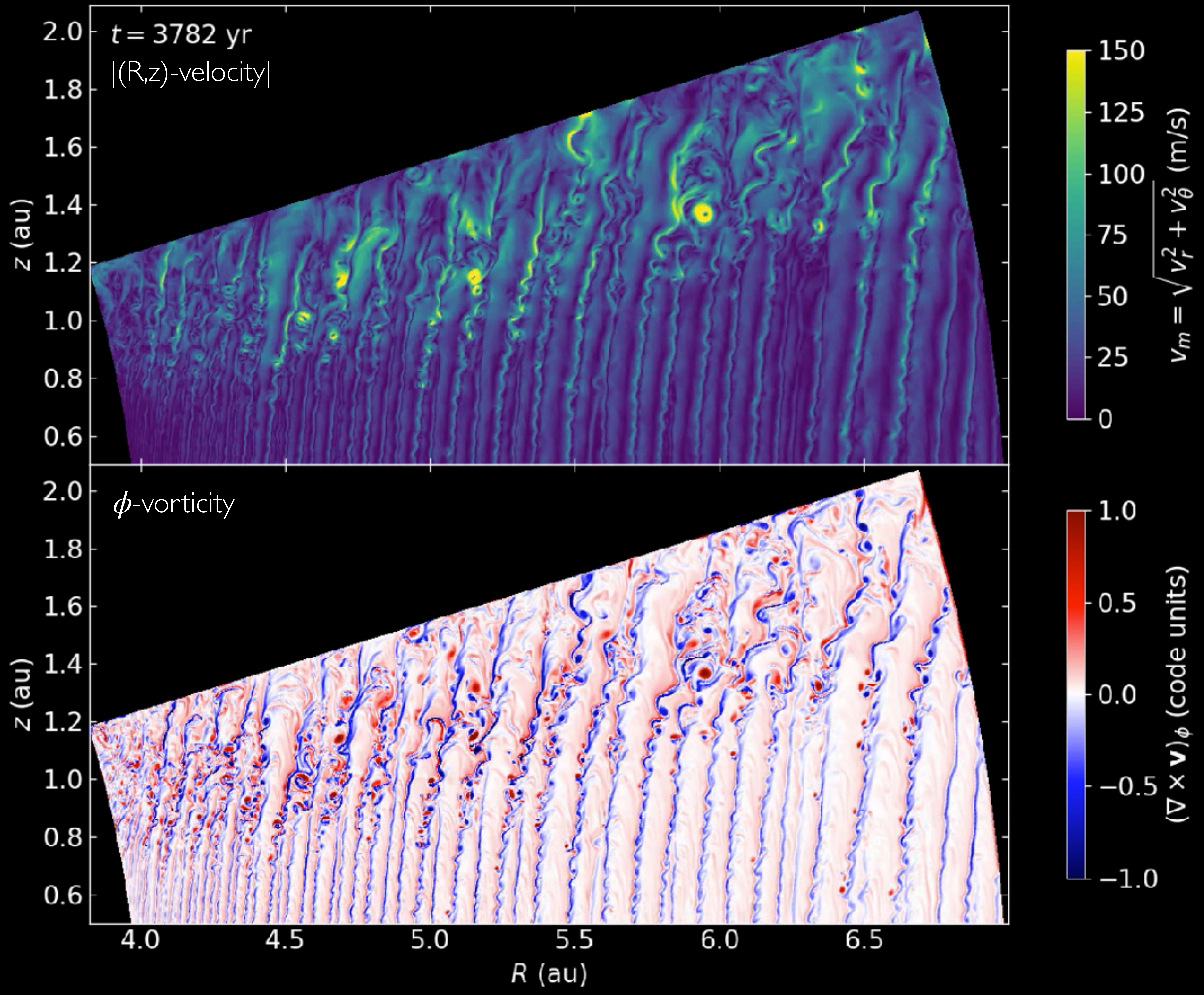
BAROCLINIC VORTEX AMPLIFICATION **(HUBERT'S TALK YESTERDAY)**



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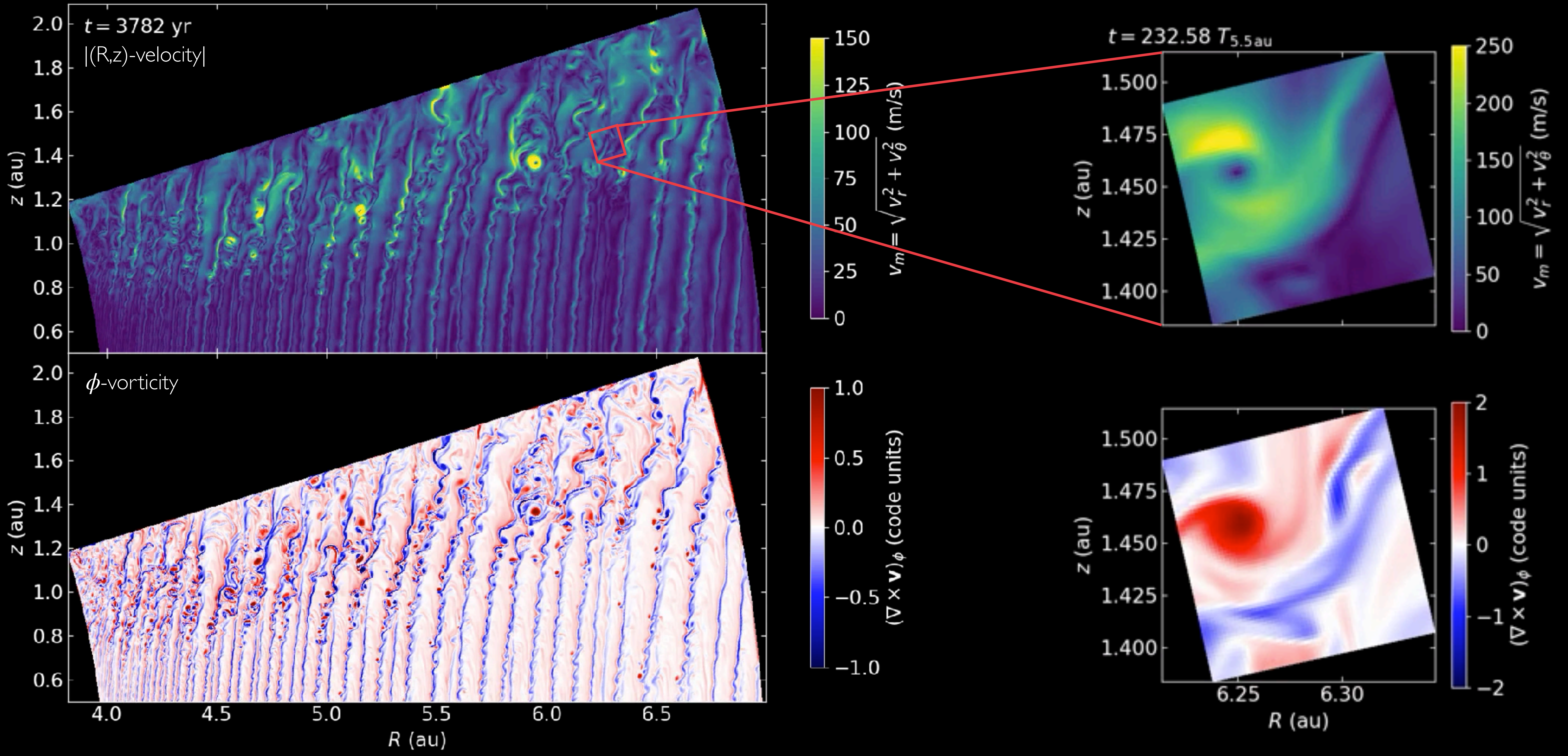
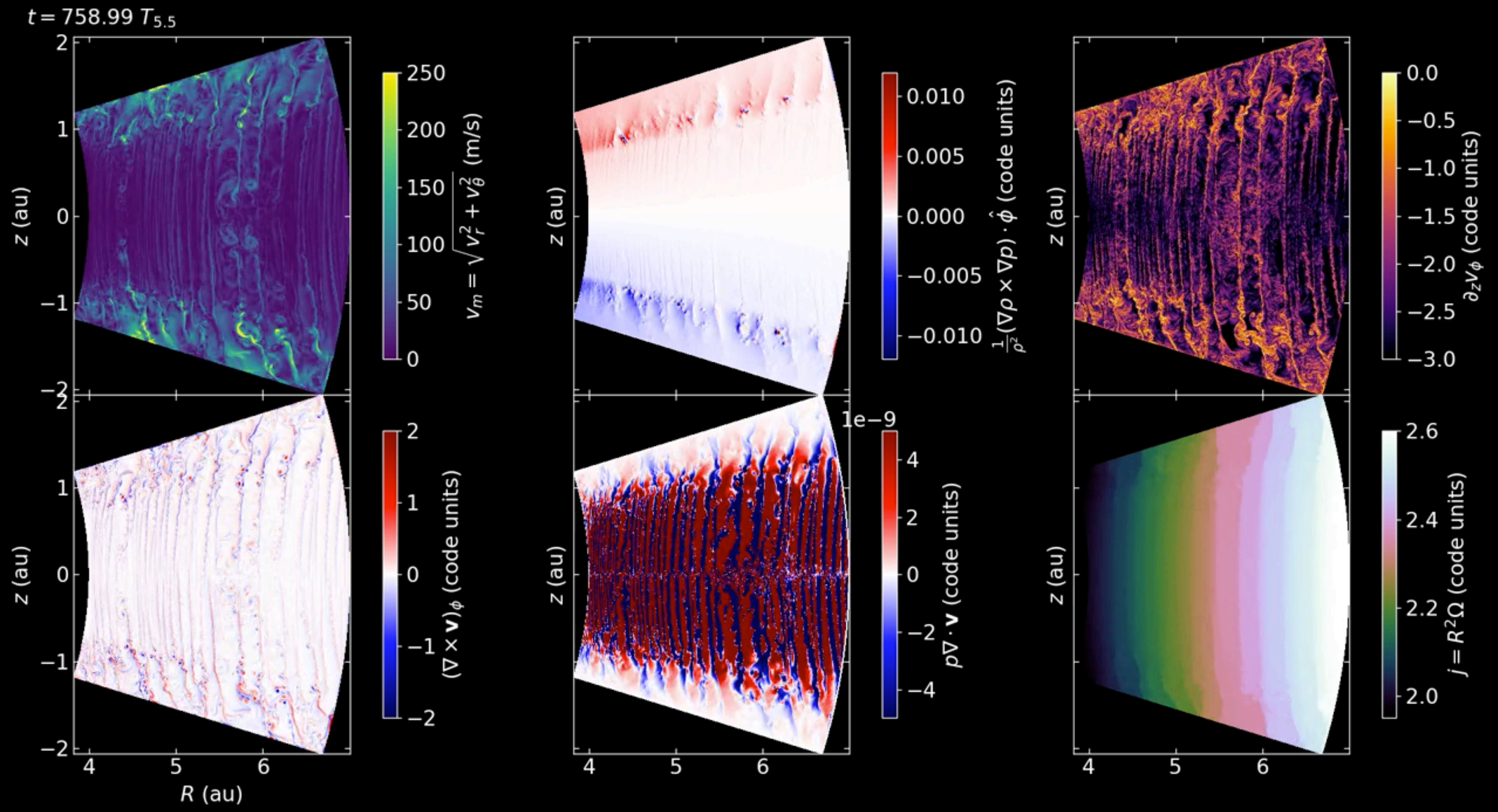
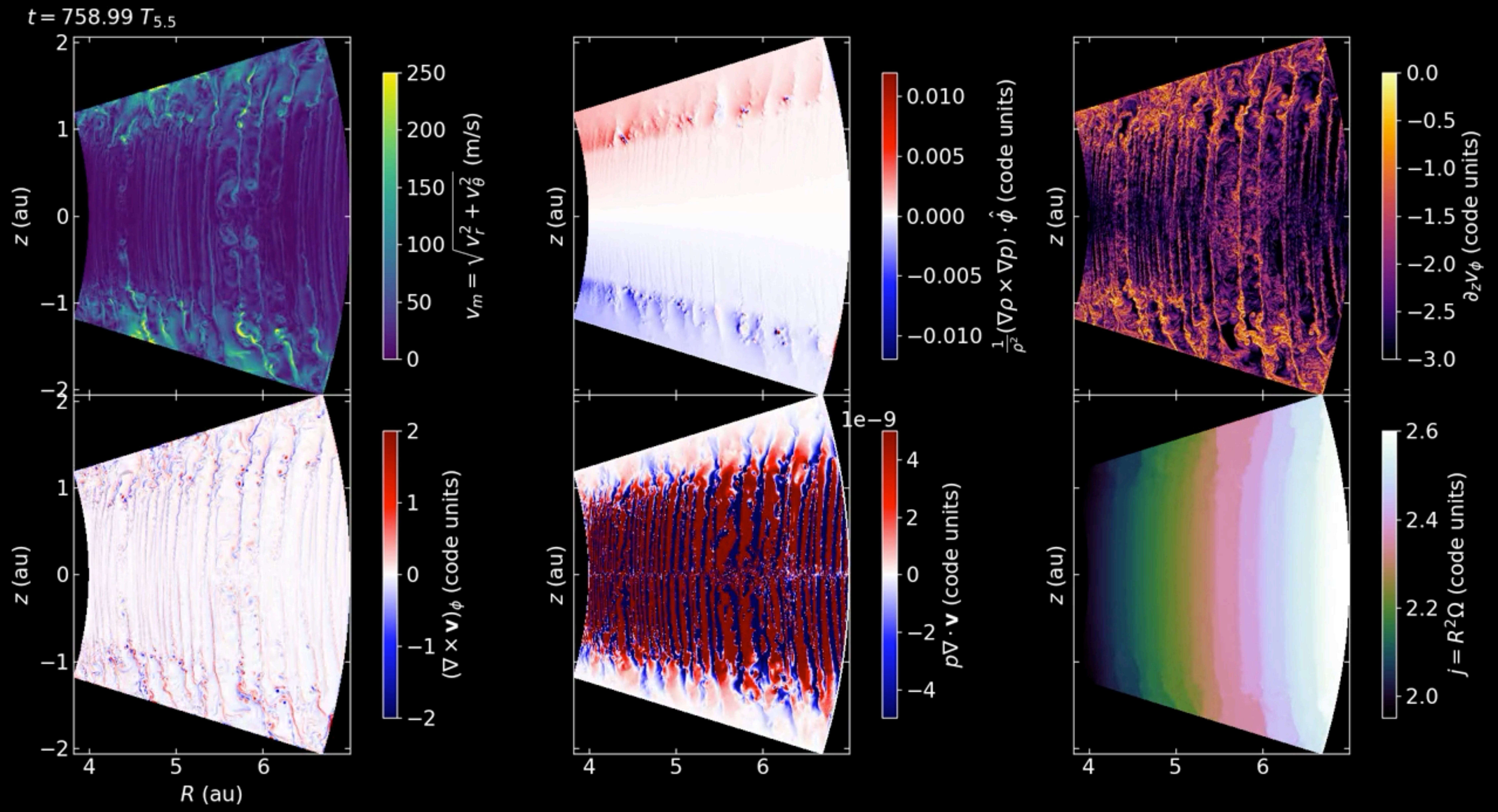


Melon Fuksman+ 2024, Klahr+ 2026



In axisymmetry, uniform-j_z bands grow only limited by the domain size

But these bands should be unstable in 3D to global non-axisymmetric modes (~ Papaloizou-Pringle instability)



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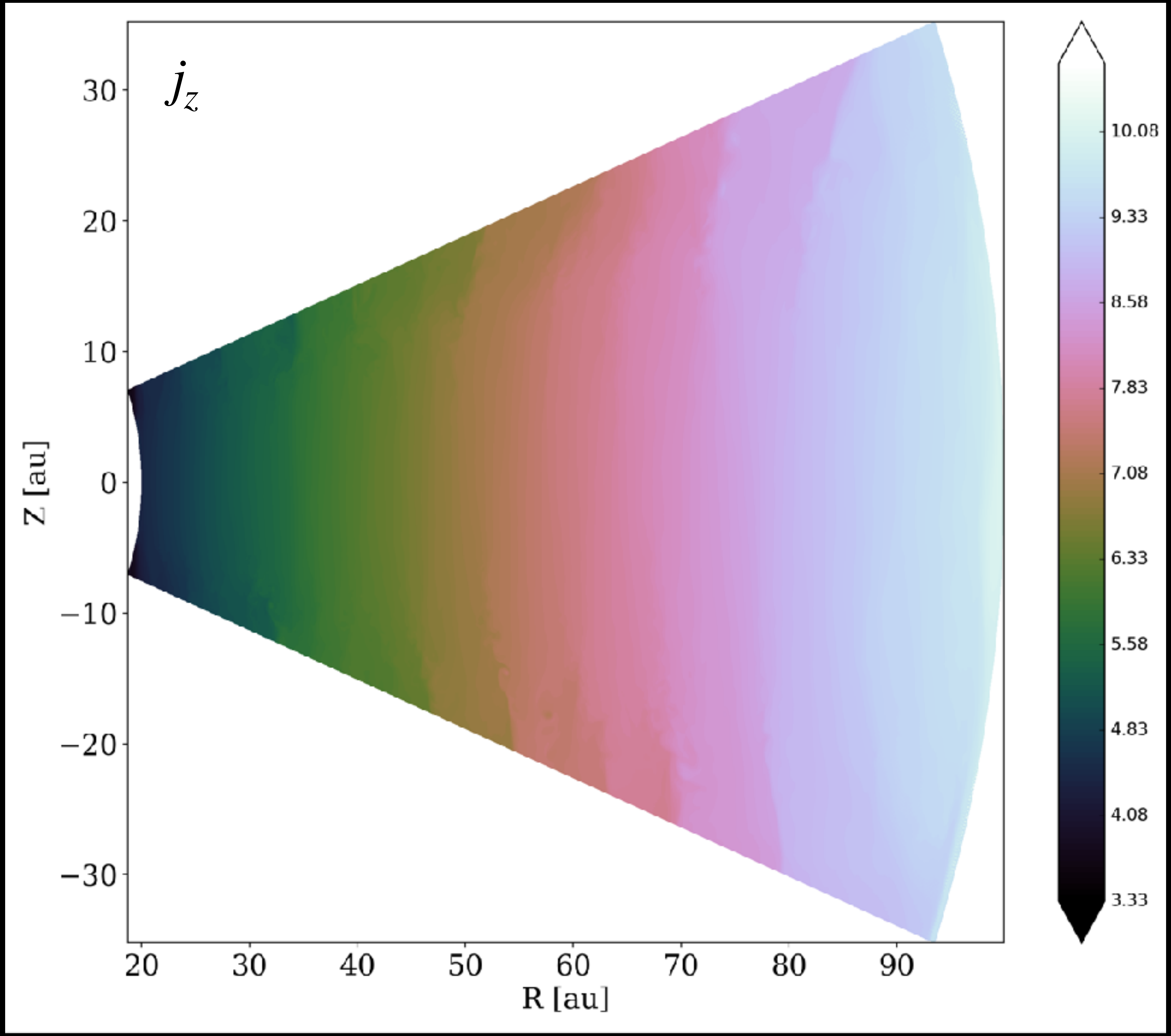
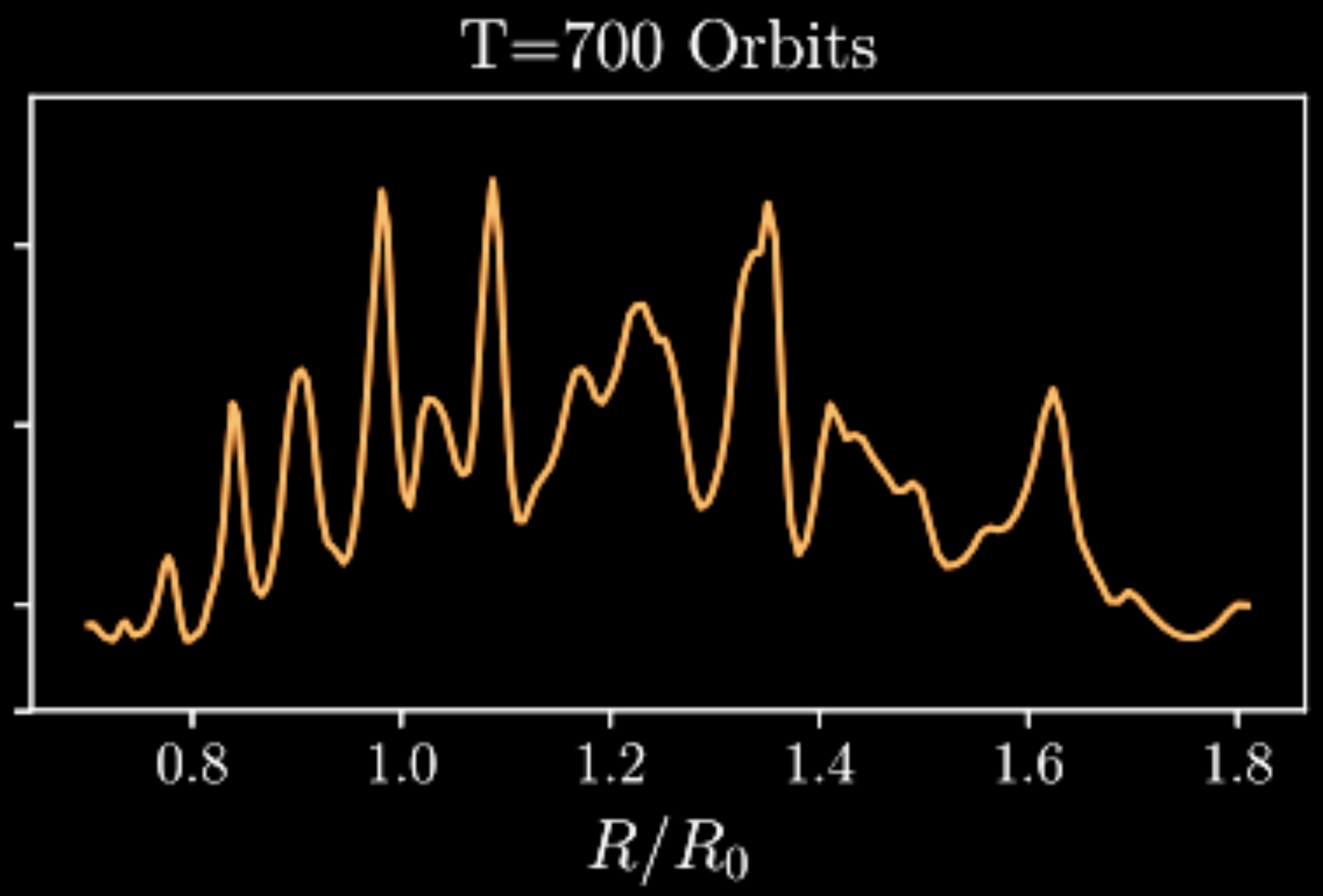
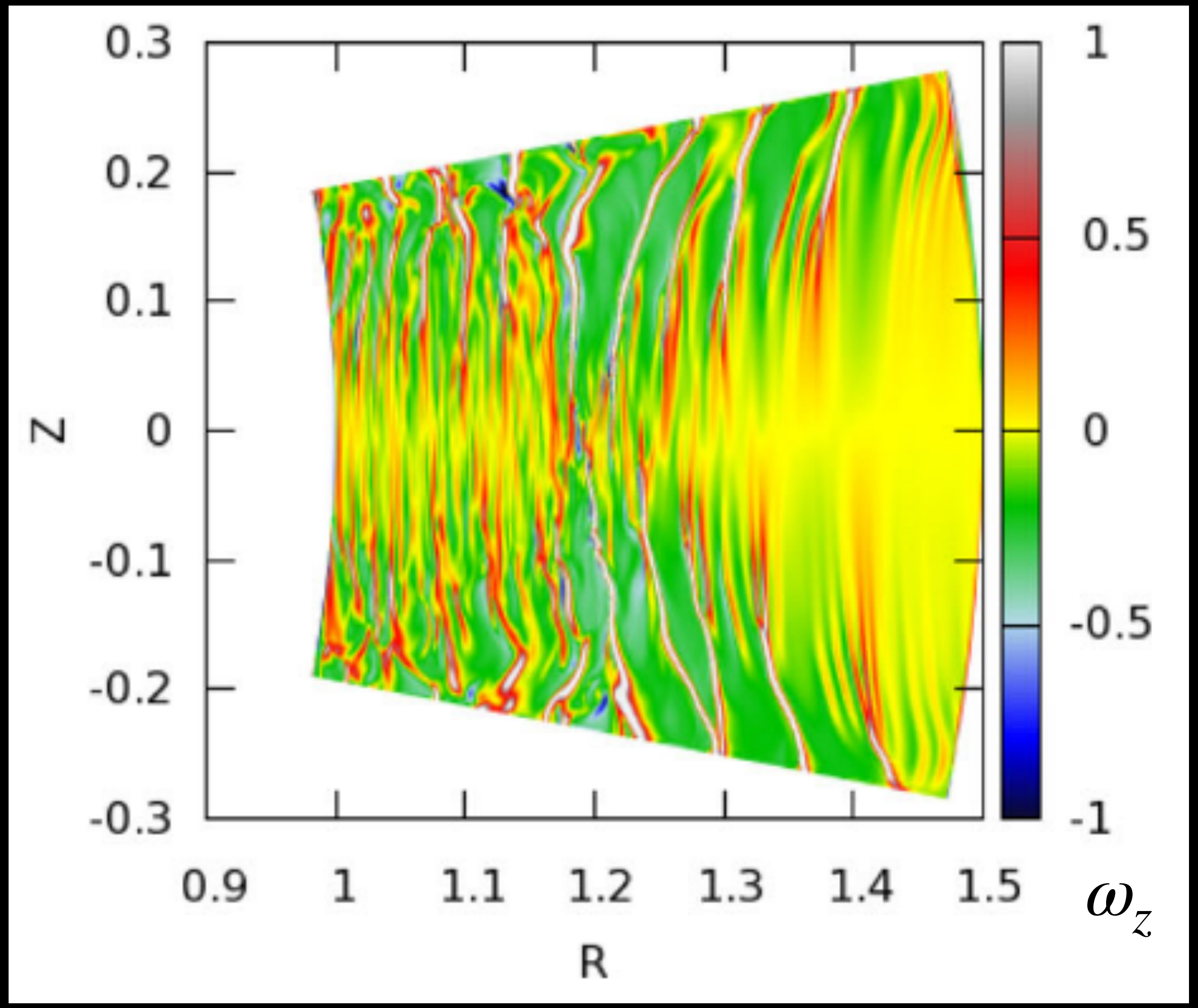
Specific angular momentum in **Flock+ 2020**



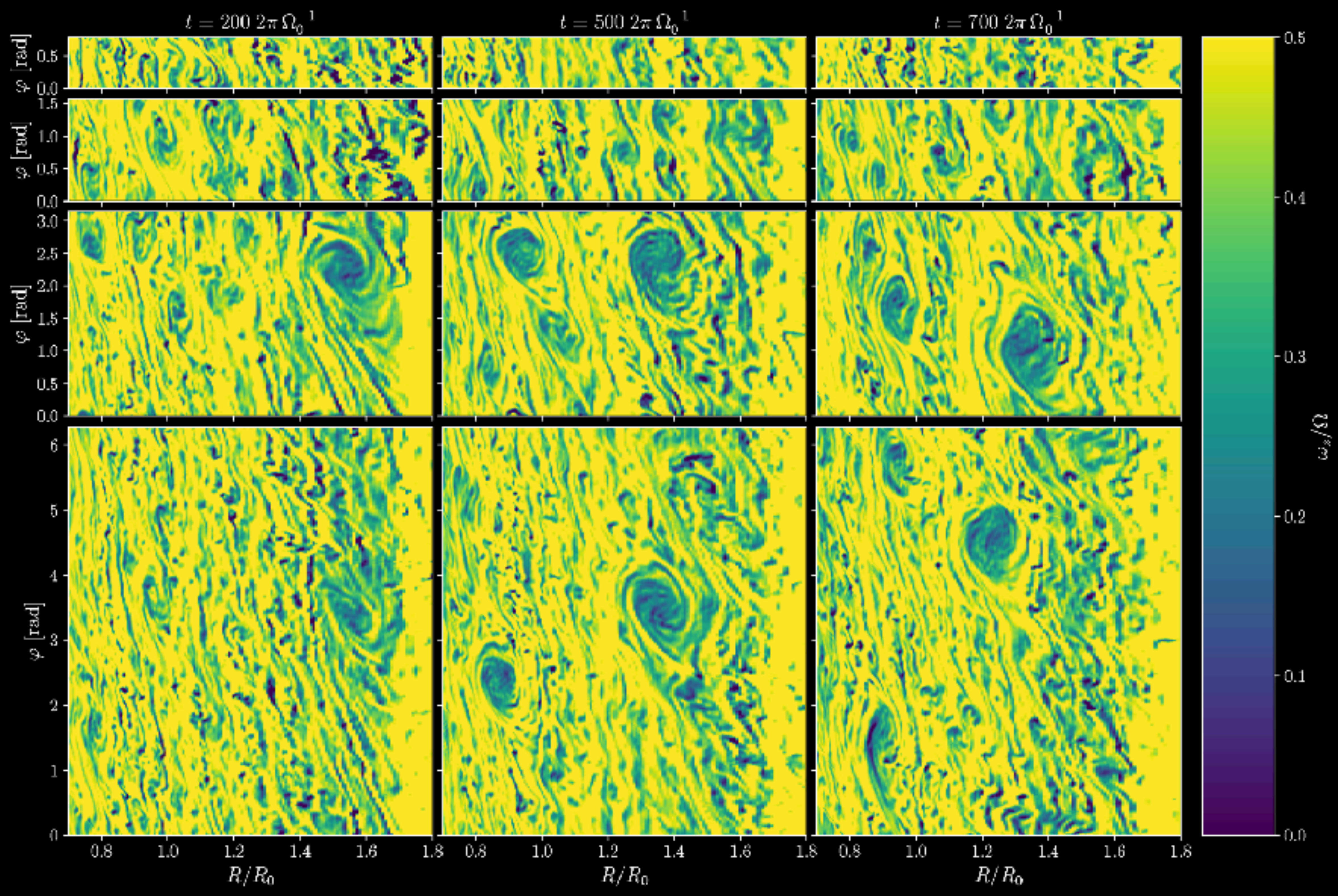


Manger+ 2018, jumps in $\mathscr{L} \propto 1/\partial_R j_z$



Richard+ 2016, reduction of vertical vorticity

- Evidence for reduced- ∇j_z bands in several published works.
- These could result from a competition between the VSI forming bands of reduced ∇j_z & non-axisymmetric modes destroying them
- The jumps in v_ϕ in between these bands might be responsible for the formation of vortices via RWI (Manger+ 2018, Pfeil+ 2021, see however Lesur & Latter 2025)



Manger+ 2018, vortices produced by Rossby Wave Instability in VSI simulations